

F . O . R . G . E  
POSTERIOR OBSERVATORY

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## Final Report

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Faculty Advisor: Professor Ken Johns (UA ATLAS HEP Group)

Due: Monday, August 10, 2026

Live observatory: [forge-observatory.com](https://forge-observatory.com)

Project documentation site: [n-herling-mk1.github.io/INFO\\_698\\_documentation](https://n-herling-mk1.github.io/INFO_698_documentation)

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## Skills Brought to the Project

This capstone draws on a cross-disciplinary foundation aligned to FORGE's three heterogeneous reproductions and its Bayesian core:

- **Bayesian deep learning** — posterior inference with Hamiltonian Monte Carlo and Last-Layer Laplace approximations; epistemic-vs.-aleatoric uncertainty quantification and calibration.
- **Machine-learning engineering** — CNN, graph, and MLP architectures in PyTorch; training, validation, and reproducible experiment pipelines.
- **High-energy physics analysis** — ATLAS Run 3 displaced-vertex long-lived-particle searches under the UA ATLAS HEP group; ROOT/CERN data-handling workflows.
- **Scientific software & deployment** — web dashboards, hosting, and continuous-integration validation across a multi-repository structure.
- **Cross-domain physical-science background** — combined Physics, Electrical & Computer Engineering, and Biology training, supporting FORGE's deliberately diverse benchmark domains.

## Revision History

Prepared by Nathan Herling

| Version | Summary of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Date     |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 0.1     | Converted hybrid (Software + Research) template to LaTeX; title page and Executive Summary drafted                                                                                                                                                                                                                                                                                                                                                                                                                                 | 06/01/26 |
| 0.2     | Executive Summary reworked to the three-experiment elevator pitch; Sections 2/2b (user/market + literature), 3/3b (features + research deliverables), and 4 (Summer 2026 milestones + embedded Gantt) filled                                                                                                                                                                                                                                                                                                                       | 06/05/26 |
| 0.3     | Proposal signed and approved; document frozen as the project's statement of work                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 06/12/26 |
| 1.0     | <b>Final Report.</b> Document re-scoped from proposal to final report: proposal Sections 1–4 carried forward unchanged; four new sections added — §5 final stats and goal verdicts, §6 research component, §7 results, §8 conclusion & future work — followed by a new reference list (§9), the proposal's ethics/approvals/appendix tail renumbered to §§10–12, and a new glossary (§13). New title-page brand lockup; three new appendices (D model roster, E frontier, F deployed site); screenshot figures for the observatory | 08/10/26 |
| 1.0     | Base measure recorded as <b>variance</b> (law of total variance) rather than entropy / mutual information, locked 08/05/26; FF1–FF4 restated as four operations on one variance budget; G1 recalibrated, G2 restated as 50 % at $2\times$ data, G3 rewritten as signed relative epistemic-variance error, G4 rewritten as an exact Euler allocation                                                                                                                                                                                | 08/10/26 |
| 1.1     | <b>Executive Summary rewritten.</b> A new opening paragraph defines aleatoric and epistemic error and states the model–data question FORGE is built to answer; the FORGE definition that opened the approved version now follows from it. §1 is therefore not carried forward verbatim; see also revision 1.2. Also: the five works cited in Table 4 added to §9, §10 forced to the head of its own page, Appendix F repointed at the live site, and a Data-science block added to the glossary                                    | 08/10/26 |
| 1.2     | <b>New §3b subsection.</b> <i>Original problem, model design and problem architectures</i> added at the end of §3b: per domain, the problem that existed before any network did, how it was posed as a machine-learning problem, and where the original statement of it is documented. The published ATLAS one-vertex analysis (Phys. Rev. D 112:092001) added to §9 and cited there. §§1 and 3b are therefore the two sections of 1–4 that are <i>not</i> carried forward verbatim; §§2 and 4 are unchanged                       | 08/10/26 |

## SPECIAL NOTE

*This is the **final report**, written as an **extension of the approved proposal** rather than as a separate document: Sections 1–4 are the proposal as approved, Sections 5–8 are new, Section 9 is the reference list, Sections 10–12 are the proposal's administrative tail, and Section 13 is a glossary. New sections are flagged in amber, and every technical term, symbol and metric is defined in the glossary. FORGE remains a **hybrid** software + research project, so Sections 2 and 3 each carry a product track and a research track (2b and 3b).*

### Word count

*The narrative body — Sections 1 through 8, excluding the front matter, the Sections 9–13 back matter per the proposal's own convention, and inline citations — runs to **10,941 words**. It divides as follows.*

|                                                      |              |
|------------------------------------------------------|--------------|
| Approved proposal, carried forward unchanged         | 4,003        |
| New material written for this report                 | <b>6,938</b> |
| §1 Executive Summary — new opening                   | 227          |
| §3b Original problem, model design and architectures | 457          |
| §5 Final project — final stats                       | 1,677        |
| §6 Research component                                | 1,906        |
| §7 Results                                           | 2,000        |
| §8 Conclusion and future work                        | 671          |

**Graded body total** **10,941**

**Rationale for exceeding the typical 5,000–6,000 words.** *This is a dual-track capstone: a deployed software product and a measured research instrument, each owing its own account, and neither legible without the other. Three experiments had to be reproduced, refined, and then measured against metrics that had themselves to be derived first. That load does not compress: each domain carries its own reproduction, instrument modifications and results, and the product they feed has to be described as well. The proposal is reproduced in full rather than summarised so commitment and outcome sit on facing pages — Section 5's verdict table is only meaningful against Sections 3 and 3b as written. A reader who has seen the proposal may skip to Section 5.*

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# 1. Executive Summary

Prepared by Nathan Herling

Machine-learning models routinely report how well they score on a held-out test set. What they rarely report is the relationship between the architecture that produced that number and the data that trained it — whether a better score is still available at all, and from which direction it would have to come. Two kinds of error sit behind that question. *Aleatoric* error is the noise carried by the data itself: label ambiguity, measurement resolution, genuine overlap between classes. It is *irreducible* — no change of architecture and no further sample will remove it. *Epistemic* error is the model’s own ignorance: the spread of predictions across every parameter setting the training data has failed to rule out. It is *reducible* — more data, better data, or a different architecture will shrink it. A single point estimate reports their sum and distinguishes neither, so a confident model and a model that has merely run out of evidence look identical on the page. FORGE turns established Bayesian machinery, together with several constructions developed for this project, on that interaction between model and data, and renders the result as a dashboard a practitioner can read at a glance and still act on.

FORGE — Functional-posterior Observatory for Refinement and Geometric Exploration — is the instrument built for that job: a web-hosted observatory around the question every trained model hides: *is its confidence earned, or has it simply run out of data?* The *long term goal* is a general platform where a researcher loads their own working model, flips it from a single point estimate to a full Bayesian posterior, and steers that uncertainty in real time — a closed-form Last-Layer Laplace Approximation (LLLA) for instant feedback, Hamiltonian Monte Carlo (HMC) for ground truth — serving two needs from one interface: exploring whether a model has reached its *epistemic limit*, and visualizing and tuning that uncertainty with Bayesian tools, with no pipeline for the user to build. The `mk_1` release reported here is the *proof of concept* for that vision, not the general platform itself. Rather than accept arbitrary user-supplied models, `mk_1` reproduces three fixed targets drawn from deliberately different sources — one published result, one student project, and one active research field — preloaded by the author with documented performance, and demonstrates the Bayesian-exploration layer on those fixed reproductions. The science and the interface are real and end-to-end; what `mk_1` deliberately scopes out is open-ended model ingestion, which a later release would add.

The motivation is a gap in everyday practice. A conventional network returns one number per input and reveals nothing about how much an equally valid version of the same model — a different seed, bootstrap, or architecture — would disagree. Bayesian inference replaces the single weight vector with a posterior  $p(w \mid D)$ , turning every prediction into a distribution whose width is the model’s *epistemic* uncertainty made visible (as distinct from *aleatoric*, irreducible noise in the data).<sup>1</sup> FORGE exposes that distinction directly in the browser and pairs it against the original deterministic baseline, lowering the barrier to interpreting model uncertainty for both specialists and reviewers. Because a posterior is computationally expensive, FORGE keeps most of the network deterministic and makes only the final layer Bayesian: the LLLA path is fast enough to resample within an interactive session, while HMC supplies the trusted reference.

The three reproductions span deliberately heterogeneous domains, to show the FORGE pattern gener-

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<sup>1</sup>The aleatoric/epistemic distinction follows A. Kendall and Y. Gal, “What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?”, NeurIPS 2017.

alizes across model classes: (1) music-genre recognition with a convolutional network and a Bayesian classification head (GTZAN, a publicly available dataset widely used for genre recognition projects); (2) phonon density-of-states regression with a graph network (Chen et al., 2021); and (3) the ATLAS Run 3 search for long-lived hidden-sector scalars via displaced muon-spectrometer vertices. Reproducing three rather than one is also a redundancy strategy: if one proves intractable on the term timeline, the proof of concept still stands on the remaining two.

## 2. User/Market research

*Prepared by Nathan Herling*

### Overall market

FORGE sits in the uncertainty-quantification (UQ) tooling space for machine learning, which has grown alongside the adoption of Bayesian methods in the data-rich physical sciences (high-energy physics, condensed matter, materials). The existing landscape splits cleanly into five segments: deterministic ML “playgrounds” (no uncertainty), post-hoc posterior diagnostics shipped as libraries, probabilistic-programming frameworks, production UQ/monitoring dashboards, and interactive sampler explainers built on toy targets. None of these is a hosted, interactive instrument for asking whether a *trained* model’s confidence is earned. That gap — between a code-heavy probabilistic framework and a static decision plot — is FORGE’s opening.

### Target users

FORGE answers two questions with one instrument, so it serves two user types. They are not redundant: the first asks a decision question with a cost attached; the second asks an open “how” question. The diagnostician is the motivating problem (what makes the work worth doing); the observatory is the capability problem (what the tool literally does, and what hypotheses H1–H3, defined in Section 3b, test). Each is framed below as User + Need + Because, with the design-actionable insight in the “Because” clause.

|                               |                                                                                                                                                                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User 1 — Diagnostician</b> | <i>“Should I collect more data, or is this as good as it gets?”</i>                                                                                                                                                                                          |
| <b>User</b>                   | Graduate researchers in data-rich physical sciences who train their own neural-net classifiers/regressors but are <i>not</i> Bayesian-ML specialists.                                                                                                        |
| <b>Need</b>                   | To see whether a trained model’s confidence is earned or borrowed — whether its remaining error is epistemic (fixable with more/better data or a different architecture) or irreducible.                                                                     |
| <b>Because</b>                | A single point estimate hides what the model does not know, and standing up a full Bayesian pipeline from scratch is enough engineering that researchers skip it and ship overconfident models. <i>This is exactly why FORGE preloads working baselines.</i> |

**Table 1:** User 1 point-of-view (the motivating, diagnostic use)

|                             |                                                                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User 2 — Observatory</b> | <i>“What is driving this uncertainty, and what happens if I change it?”</i>                                                                                                                                                                              |
| <b>User</b>                 | Practitioners and students of Bayesian deep learning who want to understand how their modeling choices shape the posterior.                                                                                                                              |
| <b>Need</b>                 | An interactive instrument to reshape the posterior in real time — turn a prior, sampler setting, likelihood, or architecture knob — and immediately see the geometry respond.                                                                            |
| <b>Because</b>              | The normal Bayesian loop is slow (minutes per HMC run) and its outputs are scalars and trace plots, not geometry you can steer. <i>This is exactly why the closed-form LLLA fast path is non-negotiable for real-time knob response (hypothesis H3).</i> |

**Table 2:** *User 2 point-of-view (the general, exploratory capability)*

### Existing competitors

Each comparable below overlaps FORGE on at least one axis (interactive model exploration, posterior/uncertainty visualization, sampler diagnostics, or Bayesian experimentation), paired with the line that separates it from FORGE.



| Comparable                                                                                                                                                                                                                                                        | How FORGE differs                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCMC Interactive Gallery; GP-regression demo (interactive sampler/Bayesian explainers)<br><a href="https://chi-feng.github.io/mcmc-demo">chi-feng.github.io/mcmc-demo</a><br><a href="https://templ.fi/gp">templ.fi/gp</a>                                        | Animate samplers on <i>fixed</i> 2-D targets or 1-D GP regression; FORGE samples a <i>trained model's</i> weight posterior across real benchmarks and adds a deterministic-vs-Bayesian comparison. |
| TensorFlow Playground; ConvNetJS (deterministic NN sandboxes)<br><a href="https://playground.tensorflow.org">playground.tensorflow.org</a><br><a href="https://cs.stanford.edu/people/karpathy/convnetjs">cs.stanford.edu/people/karpathy/convnetjs</a>           | Decision-boundary sandboxes with <i>no</i> uncertainty; FORGE adds the epistemic layer they deliberately omit.                                                                                     |
| ArviZ; bayesplot; ShinyStan (post-hoc posterior diagnostics)<br><a href="https://python.arviz.org">python.arviz.org</a><br><a href="https://mc-stan.org/bayesplot">mc-stan.org/bayesplot</a><br><a href="https://mc-stan.org/shinystan">mc-stan.org/shinystan</a> | Plot samples you <i>already</i> have; FORGE is hosted and generates and reshapes posteriors in-session.                                                                                            |
| PyMC; Pyro; JASP (probabilistic-programming / stats)<br><a href="https://pymc.io">pymc.io</a><br><a href="https://pyro.ai">pyro.ai</a><br><a href="https://jasp-stats.org">jasp-stats.org</a>                                                                     | Code frameworks or a stats package; FORGE is the no-code “turn a knob, watch the posterior move” layer on top.                                                                                     |
| Weights & Biases; WhyLabs; Evidently AI (production UQ/monitoring)<br><a href="https://wandb.ai">wandb.ai</a><br><a href="https://whylabs.ai">whylabs.ai</a><br><a href="https://evidentlyai.com">evidentlyai.com</a>                                             | Track deployed-model metrics and drift; report confidence numbers, not posteriors or Bayesian reasoning.                                                                                           |
| Netron (architecture viewer)<br><a href="https://netron.app">netron.app</a>                                                                                                                                                                                       | Shows structure only — no uncertainty, posterior inference, or Bayesian reasoning.                                                                                                                 |

**Table 3:** Comparable tools and how FORGE is differentiated

## User insights and pain points

Three pain points recur across both user types and drive the architecture directly: (i) the engineering cost of standing up a Bayesian pipeline (HMC/NUTS, calibration, last-layer Laplace) is high enough that researchers skip it — addressed by preloading working baselines that users inherit rather than rebuild; (ii) the edit-code-and-wait loop is slow and its outputs are scalars and trace plots rather than steerable geometry — addressed by the closed-form LLLA fast path that keeps the knob panel responsive, with heavy HMC reserved for ground truth; and (iii) the elephant in the room for any trained model is whether it will *generalize* and — more pointedly — how anyone would know, yet practitioners have no lightweight, hands-on way to interrogate it: to *play* with the constraints of the model and its data and watch the epistemic limit emerge, separating confidence that is earned from confidence that is merely extrapolated, whether out of plain curiosity or as fact-finding with real-world stakes in research and industry — addressed by FORGE’s interactive exploration layer, which surfaces epistemic uncertainty

(the posterior width where the data stops pinning the model down) as a visible, steerable proxy for that question rather than something a user must re-engineer from scratch. In each case the problem framing and the technical design are the same claim seen from two sides.

## 2b. Literature Review/Market research

*Prepared by Nathan Herling*

Because FORGE has both a software and a research element, novelty is gauged on two fronts: the comparable-tools scan above (so the build is clear of others’ ground) and the “research of research” below — peer-reviewed work that grounds FORGE’s ingredients and sharpens where its contribution is genuinely new. A continually-updated, browsable version of this review is maintained as the [Literature Search](#) panel on the project documentation site.

### Methodological foundations

Each FORGE ingredient rests on established work. Bayesian deep-learning inference: variational Bayes-by-backprop (Blundell et al., 2015) and Monte-Carlo dropout as approximate inference (Gal & Ghahramani, 2016). The interactive fast path: the Last-Layer Laplace Approximation and its “Bayesian last layer” framing (Kristiadi et al., 2020; Daxberger et al., 2021, *laplace-torch*). The ground-truth path: Hamiltonian Monte Carlo and the No-U-Turn Sampler (Neal, 2011; Hoffman & Gelman, 2014). The aleatoric/epistemic decomposition that motivates the whole tool: Kendall & Gal (2017). The reproduced experiments draw on domain baselines: a graph network for phonon DOS (Chen et al., 2021) over the DFPT phonon database (Petretto et al., 2018), and CNN-based genre recognition (Sridhar, 2024; and the author’s prior BEARDOWN work, Herling & Attili, 2025).

### Closest prior art and the novelty gap

Table 4 collects the work nearest to FORGE’s *purpose*—a posterior over models used to decide where knowledge is lacking—and its *interactivity*.

| Reference                                                                                                                 | Relevance to FORGE                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dearden, Friedman & Andre — <i>Model-Based Bayesian Exploration</i> (UAI 1999)                                            | An early statement of the core FORGE logic — a posterior over models beats a point estimate for deciding where knowledge is lacking — but applied as an RL exploration algorithm, not a visualization tool.                    |
| Frazier — <i>A Tutorial on Bayesian Optimization</i> (2018)                                                               | Shares the “visualize posterior uncertainty, decide where you’re unsure” pattern, but aimed at optimizing a function rather than diagnosing a trained model.                                                                   |
| Wang, Jin, Schmitt & Olhofer — <i>Recent Advances in Bayesian Optimization</i> (ACM CSUR 2023)                            | Confirms the BO field is optimization-centric; useful for positioning FORGE’s model- <i>diagnosis</i> purpose as distinct.                                                                                                     |
| Gabry, Simpson, Vehtari, Betancourt & Gelman — <i>Visualization in Bayesian Workflow</i> (JRSS-A 2019)                    | The methodological backbone for FORGE’s plots — but static, R-package workflow guidance, not an interactive hosted dashboard.                                                                                                  |
| Nussbaumer, Böck & Cito — <i>Online and Interactive Bayesian Inference Debugging</i> (ICSE 2026; <i>InferLog Holmes</i> ) | The closest prior art to FORGE’s online, interactive ambition — but an IDE debugger for PPL code aimed at <i>fixing inference</i> , not a hosted multi-domain dashboard for exploring a trained model’s epistemic uncertainty. |

**Table 4:** Research landscape — closest prior art

### Where FORGE is novel

The landscape grounds each ingredient individually, but no existing system is a hosted dashboard that simultaneously (1) preloads trained models across heterogeneous domains, (2) toggles deterministic ↔ Bayesian and compares posteriors against the baseline, and (3) reshapes posterior geometry in real time through a knob panel backed by closed-form LLLA plus heavy HMC. The combination, framed by the question “*is this model’s confidence earned or borrowed—has it reached its epistemic limit?*”, is the differentiated contribution.

## 3. Product Features

*Prepared by Nathan Herling*

FORGE is built in three maturity rungs of *one* product, not three different products: the same application (data locked, methodology exposed) is carried up a ladder — first proven locally (*MVP — phase 1*), then deployed (*MVP — deployed*), then optionally opened to an LLM agent (*Stretch — MCP*). The science underneath all three is identical, so the project always has a complete deliverable no matter how high the ladder is climbed. Phase 1 (features F1–F5) is everything required to demonstrate the concept locally; the deployed rung (F6) ships the same stack to a cloud host; only the MCP rung (S1) is a true stretch, and it adds no new science. All features are owned by Nathan Herling (solo capstone).

| #                                                           | Feature                                         | What it does / why the user wants it                                                                                                             |
|-------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MVP — phase 1</b> (minimum feature set; local + logging) |                                                 |                                                                                                                                                  |
| F1                                                          | Preloaded multi-domain baselines                | Three working, documented models (genre, phonon, ATLAS) load instantly; the user inherits a baseline instead of building a pipeline.             |
| F2                                                          | Deterministic $\leftrightarrow$ Bayesian toggle | Flip a trained model from a point estimate to a posterior and compare the two side by side — the core “is the confidence earned?” view.          |
| F3                                                          | Real-time knob panel                            | Turn prior, sampler, likelihood, and architecture knobs and watch the posterior geometry respond live, backed by the closed-form LLLA fast path. |
| F4                                                          | Posterior-geometry visualization                | Credible regions, function-space spread, and the per-config posterior- $\sigma$ distribution, rendered in-browser.                               |
| F5                                                          | Resource-logging layer                          | Per-run telemetry (memory, time, cost) that turns Phase-2 infrastructure sizing into an empirical fit rather than a guess.                       |
| <b>MVP — deployed</b> (same stack, cloud host)              |                                                 |                                                                                                                                                  |
| F6                                                          | Cloud deployment                                | The same Docker images run on a cloud host; only one environment variable changes between local and deployed.                                    |
| <b>Stretch — MCP</b> (optional agent interface)             |                                                 |                                                                                                                                                  |
| S1                                                          | MCP server                                      | A thin manifest advertises the compute functions as tools an LLM client (e.g. Claude Desktop) can call — no new infrastructure.                  |

**Table 5:** FORGE feature list across the three MVP rungs, all owned by N. Herling

### Feature F3: Real-time knob panel — the load-bearing interactive feature

The whole value of FORGE for User 2 depends on the knob panel feeling instant, which is why the architecture splits into two compute paths: a closed-form LLLA path on an always-on CPU for interactivity, and a heavy HMC/NUTS path on a serverless GPU for ground truth. The performance envelope is summarized in Table 6.

| Parameter                   | Target                | Ceiling               | Comments                                                |
|-----------------------------|-----------------------|-----------------------|---------------------------------------------------------|
| LLLA knob-to-update latency | $\leq 1\text{ s}^*$   | $\leq 3\text{ s}^*$   | Closed-form last-layer; must feel interactive           |
| HMC “ground-truth” resample | $\sim 1\text{ min}^*$ | $\leq 5\text{ min}^*$ | Serverless GPU, scale-to-zero; per-job, not interactive |
| Compute paths exposed       | 2                     | 2                     | LLLA (CPU, always-on) and HMC/NUTS (GPU, on demand)     |

*\* Early planning estimates, not yet profiled — to be revisited once Phase 1 and its resource-logging layer (F5) are complete and the fitted time-vs-scale curves ( $\text{time} \approx f(n_{\text{params}}, n_{\text{samples}}, \dots)$ , including data size) are available. (Cold-start container spin-up and JAX/XLA compilation on the serverless GPU are not yet included in the HMC figures.)*

**Table 6:** Feature F3 interactive-path parameters

## Feature F6: Cloud deployment (MVP — deployed) — hosting envelope

Deployment must hold to a hobby-scale budget; the only component that must run continuously is one always-on CPU box, with the heavy GPU path serverless and effectively \$0 at idle. Planning estimates (as of May 2026) are in Table 7.

| Layer                                  | Est. cost/mo  | Notes                                        |
|----------------------------------------|---------------|----------------------------------------------|
| Frontend (static, GitHub Pages)        | ~\$1          | Domain only; Pages + TLS free                |
| Backend API + LLLA (Fly.io/Render)     | \$10–25       | Always-on CPU box, ~2 GB RAM                 |
| HMC heavy jobs (Modal, serverless GPU) | ~\$0 baseline | Scale-to-zero, per-second; credits cover dev |
| Database                               | \$0           | Stateless / explicit job-ID design; deferred |
| <b>Target total</b>                    | <b>~\$25</b>  | Size of one always-on CPU box                |

**Table 7:** Feature F6 hosting-cost envelope

## Per-project interactive demos (the “fidgets”)

Each reproduction ships a hands-on demo built into its own *standalone full stack* — a local Python Flask dev server during Phase 1 — so every project is independently runnable before Phase 2 folds them onto the shared deployed backend (these are the Phase-1 standalone sites of Section 4). All three follow one pattern: *drop or tune a domain object* → *a scored output, an uncertainty band, and a themed visualization*, over the same `/api/predict` contract used by the `/api/eda` route. They are the user-facing realization of features F1–F4, specialized per domain; the uncertainty band is the through-line that ties them to FORGE’s Bayesian spine.

| Project | Interaction                                                                 | Output (scored + uncertainty + viz)                                                                                                                                                        | Backing   |
|---------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Genre   | Drop an audio file                                                          | Per-genre softmax scores + audio-settings dashboard                                                                                                                                        | real      |
| Phonon  | Build a crystal (elements → formula / unit cell)                            | Predicted phonon DOS with a posterior ribbon; live specific heat $C_v$ and vibrational entropy on a temperature sweep                                                                      | surrogate |
| ATLAS   | Tune kinematic features (sliders, anchored to <code>eda_stats.json</code> ) | A point on the ABCD plane with <i>dual</i> NN1/NN2 credible bands off the axes; barrel/endcap toggle; in-distribution vs extrapolation shading; LLLA live with a precomputed HMC reference | mock      |

**Table 8:** Per-project interactive demos. The ATLAS payoff view is the barrel region where the NN1 and NN2 bands lock together despite the measured  $dCor \approx 1.79$  correlation failure.

Honesty caveat: only the genre demo is backed by a trained model (BEARDOWN); the phonon and ATLAS demos run against a surrogate / mock posterior until their real models land, and are labelled *demo prediction* in-UI so trained and placeholder output are never confused.

## 3b. Research Project Deliverables

*Prepared by Nathan Herling*

### Hypotheses (H1–H3)

FORGE’s observatory capability rests on three testable hypotheses, each tied to a feature and checked against the HMC/NUTS ground truth:

- **H1 — visualization fidelity.** The Bayesian posterior can be rendered in-browser (credible regions, function-space spread, and the per-configuration posterior- $\sigma$  distribution; feature F4) faithfully enough that a viewer reads the model’s epistemic uncertainty correctly rather than an artifact of the display. *Tested by* comparing the rendered credible regions and their coverage against the ground-truth posterior.
- **H2 — sampler correctness and efficiency.** The closed-form LLLA fast path agrees with the ground-truth HMC/NUTS posterior within tolerance (correctness) while staying fast enough to re-sample inside an interactive session (efficiency; the F3 latency envelope). *Tested by* LLLA-vs-HMC agreement on posterior moments and interval coverage, alongside the measured knob-to-update latency.
- **H3 — hyperparameter → posterior geometry (the keystone).** Turning the prior, sampler, likelihood, and architecture knobs produces systematic, interpretable, *steerable* changes in posterior geometry — the mapping is legible enough that a user can deliberately steer uncertainty in real time. This is the project’s central claim and the reason the LLLA fast path is non-negotiable. *Tested by* demonstrating reproducible, interpretable posterior-geometry responses to knob changes across the three reproductions.

### Final presentation format

The project culminates in (i) a written report with introduction, methods, and results sections and at least five peer-reviewed citations, carrying at least two informative graphics (the deterministic-vs-Bayesian comparison and the fitted resource-cost curves); (ii) a poster and live demo for the project presentation; and (iii) the deployed FORGE site plus three public GitHub repositories (documentation, experiments, software), tagged v1.0 at freeze.

### What analysis is being run

Three reproductions, each driven to a functional ground-truth result before the next begins, then extended with Bayesian inference:

- **Genre recognition** — a CNN with a Bayesian classification head over GTZAN, a publicly available dataset widely used for genre recognition projects (spectrogram + engineered-feature encoders).
- **Phonon DOS** — a graph network with a Bayesian last-layer head over the 51-point DOS output; the published baseline gives point predictions only, so calibrated credible intervals are FORGE’s added contribution.
- **ATLAS LLP** — a deterministic MLP discriminating signal (HSS → LLP) from QCD background, extended with HMC over the output layer.

Across all three, the analysis is posterior inference via the closed-form LLLA fast path and ground-truth HMC/NUTS, evaluated for calibration coverage in addition to raw accuracy.

Each reproduction ships between one and three trained models to the FORGE site, so the platform can showcase *differences between models*, not merely a single model’s uncertainty:

- **(1) Faithful reproduction (required).** A model reproduced to the source work’s reported configuration and performance — the fidelity anchor, present for every experiment.
- **(2) Author-optimized variant (target).** A model selected and tuned with the reliability-aware Bayesian metric of Appendix C, which scores candidates on accuracy, posterior uncertainty, and cross-validation stability rather than accuracy alone — the model on which FORGE’s epistemic-selection idea is exercised in practice.
- **(3) Data-permitting variants (stretch).** Where the dataset supports it, additional flavors of (2) that vary the input regime — e.g. for genre recognition, parallel 30-second and 3-second-clip models whose posteriors can be compared head-to-head.

Model (1) is a hard deliverable for all three reproductions; (2) is the goal wherever the term timeline allows; (3) is opportunistic.

### **What accuracy is expected**

Each reproduction commits up front to targets drawn from its source work — FORGE’s “empirical contract” for declaring a baseline reproduced. Calibration coverage is the additional contribution layered on top. These targets govern the faithful reproduction (model 1); the author-optimized variant (model 2) is judged instead by the reliability-aware composite of Appendix C, not by beating the fidelity target.

| Experiment        | Reproduction target                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Genre recognition | Best-config validation accuracy $\geq 0.75$ (cfg_14: 0.766); sweep mean $\approx 0.72$ ; per-genre F1 in 0.46–0.89 (BEARDOWN, Herling & Attili 2025).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Phonon DOS        | $\geq 70\%$ of test materials within 10% relative error on the average phonon frequency; per-element DOS MSE $10^{-3}$ – $10^{-1}$ (Chen et al. 2021).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ATLAS LLP         | <ul style="list-style-type: none"> <li>• <b>Reproduce the baseline (Schott 2024<sup>2</sup>)</b>: matched two-hidden-layer 128–128 MLPs — ReLU hidden, sigmoid output; binary-crossentropy loss, Adam; 10 epochs, batch 32; 75/25 split with signal and background reweighted to equal summed weight; uncorrelated feature sets (NN1 isolation/activity, NN2 spatial distribution).</li> <li>• <b>Acceptance</b>: NN1 train/test ROC–AUC agreement, barrel and endcap (overtraining check); NN1/NN2 plane Pearson correlation <math>\approx 1\%</math> (barrel), 2% (endcap), preserving the ABCD axis-independence requirement; A-region thresholds NN1, NN2 <math>&gt; 0.5</math> (barrel), <math>&gt; 0.8</math> (endcap).</li> <li>• <b>NN2 gap</b>: no standalone ROC/AUC in the source — only the joint-plane correlation constrains its discriminating power.</li> <li>• <b>Extend (new)</b>: beyond the faithful reproduction, add architectures from recent results (Herling, Mar. 2026), selected with the Appendix C reliability-aware metric — supplying the model (2)/(3) variants and a standalone calibrated NN2 the baseline lacks.</li> </ul> |

**Table 9:** Per-experiment reproduction targets

### What if the analysis doesn’t work

A null result is acceptable and, in this project, often *informative*: a model shown to sit at its epistemic limit is exactly the diagnostic FORGE exists to surface. Risk is further managed by redundancy — three reproductions mean the platform still demonstrates the concept if one proves intractable — and by a dedicated three-week risk-management window (Weeks 8–10) that absorbs slippage from Phase 1 or Phase 2. The descope order is fixed: drop the MCP stretch (S1) first, then trim the depth of the Bayesian/visualization refinements; the three baseline reproductions (the guaranteed floor) and the write-up are never cut, because they carry the deliverable.

### What if the data isn’t available

There is no scraping dependency. All three datasets are public or already in hand: GTZAN (genre; a publicly available dataset widely used for genre recognition projects); the DFPT phonon database via the Materials Project (phonon, Petretto et al. 2018); and the ATLAS reproduction inputs from the author’s prior thesis work and Monte-Carlo samples. The C0–C3 calibration tiers are analytic, so their ground truth is always available for the CI scoring path regardless of external data.

<sup>2</sup>M. L. Schott, *Search for Long-Lived Particles Decaying into Displaced Hadronic Jets in the Muon Spectrometer in  $pp$  Collisions at  $\sqrt{s} = 13$  TeV with the ATLAS Detector*, Ph.D. dissertation, Department of Physics, The University of Arizona, 2024.



## Original problem, model design and problem architectures

Each model comes from an original question and a machine-learning architecture designed to use the available data in order to answer a research question. Set out below, per domain: the original problem, how it was posed as a machine-learning problem, and where the original statement of it is documented. The reproduction targets of Table 9 follow directly from these sources.

**Music genre — a human category with no physical observable.** The original problem belongs to music information retrieval: assign a genre label to a recording when genre is a listener’s category rather than a measurable property of the signal, so the boundaries between neighbouring labels are genuinely contested and part of the error is irreducible by construction. GTZAN (a publicly available dataset widely used for genre recognition projects) is a dataset created by Tzanetakis and Cook (2002). It includes 1000 music excerpts of 30 seconds duration, with 100 examples in each of 10 different music genres: Blues, Classical, Country, Disco, Hip-hop, Jazz, Metal, Pop, Reggae, and Rock. The adaptation to machine learning takes two views of the same clip: the waveform rendered as a mel spectrogram, which turns a time-series problem into an image problem a convolutional network can read, and a vector of 57 engineered summary statistics over the same audio. A CNN branch and a tabular MLP branch are fused and a softmax over  $K = 10$  classes returns a distribution rather than a label, which is what makes the class axis available as FORGE’s resolving axis (§6.3). The CNN methodology follows Sridhar (2024); the implementation reproduced here is the author’s own prior work (Herling and Attili, 2025).

**Phonon DOS — a calculation too expensive to run everywhere.** The original problem is condensed-matter: a crystal’s phonon density of states fixes its thermal behaviour — heat capacity, vibrational entropy, zero-point energy all follow from the spectrum — but obtaining it from first principles means density-functional perturbation theory, which is costly enough that coverage of materials space, not accuracy on any one material, is the binding constraint. Petretto *et al.* (2018) answered that with a high-throughput DFPT database; Chen *et al.* (2021) asked whether the spectrum can instead be predicted from structure alone. The adaptation is a graph problem: atoms become nodes, neighbours within a cutoff radius become edges, and the network’s convolutions are made E(3)-equivariant so that rotating, translating or reflecting the input crystal transforms the output correctly by construction instead of having to be learned from data. The prediction is a regression onto a 51-bin spectrum under mean-squared error, accepted on the fraction of test materials whose mean phonon frequency lands within 10 % of truth. That equivariance is also why the network ends in a contraction against spherical harmonics rather than a weight matrix — the architectural fact that forces the readout head of §5a.

**ATLAS LLP — separating signal from punch-through, and estimating what cannot be simulated.** The original problem is to separate a new-physics signal from jet punch-through: hadronic and electromagnetic showers not fully contained in the calorimeter leak into the muon spectrometer and reconstruct as a displaced vertex, which is exactly what a long-lived particle decay would look like (ATLAS Collaboration, 2025). That background cannot be simulated reliably, so it is estimated from the data itself by the ABCD method: two discriminants cut the plane into four regions, and the count in the signal region A is predicted from the three control regions as  $N_A = N_B N_C / N_D$  — valid only if the two discriminants are uncorrelated for background. The adaptation to machine learning is what supplies those two axes. Two deep networks are trained on deliberately disjoint input features — NN1 on vertex isolation and detector activity, NN2 on the spatial distribution of the vertex itself — each a two-

hidden-layer 128–128–1 perceptron with ReLU hidden units and a sigmoid output, trained under binary cross-entropy with Adam. Each network’s predicted probability then becomes one variable of the ABCD plane. The quantity that decides whether the arrangement worked is *closure*: how closely the estimated background agrees with the observed background in samples where the answer is already known. The published one-vertex analysis (ATLAS Collaboration, 2025) is the source for the architecture, the feature split and the region thresholds reproduced in Table 9; the local implementation this reproduction works from is Schott’s thesis. Closure is where §6.4’s contribution enters — the DNCn\_QN term penalises the closure error directly rather than the global correlation that only proxies for it.

## 4. Project Timeline & Gantt Chart

*Prepared by Nathan Herling*

The thirteen-week schedule (single contributor) is anchored at Week 0 = May 18, 2026, and is organized around the three MVP phases. Weeks 0–3 cover intro, planning, and setup (proposal mk\_1 at W2, full proposal at W3). **Phase 1** (Weeks 4–6) stands up the three reproductions one per week — genre recognition (W4), phonon DOS (W5), and the ATLAS LLP search (W6) — each as a functional standalone site driven to ground truth, with the minimum Bayesian options (LLLA + HMC), a local dashboard, and built-in compute-recording metrics; having all three functional by Week 6 is the guaranteed “floor” deliverable. **Phase 2** (Weeks 7–9) ties the three repositories into one deployed FORGE website, moves off localhost to external hosting, tests and debugs, and finalizes the Bayesian-tuning and dashboard decisions; the write-up begins in Week 9. **Phase 3 / stretch** (Week 10) locks the MVP and freezes the repositories, attempting the MCP stretch only if time allows. A dedicated **three-week risk-management window (Weeks 8–10)** absorbs slippage before the write-up (W11) and the short hand-in week (W12). Table 10 lists the dated milestones; Figure 1 is the corresponding Gantt.

| Milestone                                                                       | Date    |
|---------------------------------------------------------------------------------|---------|
| Course orientation; three repositories bootstrapped (course)                    | 5/22/26 |
| Project Proposal mk_1 submitted (course)                                        | 6/5/26  |
| Full proposal; timeline + Gantt locked (course)                                 | 6/12/26 |
| Music Genre standalone site functional (ground truth)                           | 6/19/26 |
| Phonon DOS standalone site functional (ground truth)                            | 6/26/26 |
| <b>ATLAS LLP standalone site functional — all three at ground truth (floor)</b> | 7/3/26  |
| Repos integrated into FORGE website; deployed to external hosting (Phase 2)     | 7/17/26 |
| Software feature-frozen; write-up begun                                         | 7/24/26 |
| MVP + repos locked; MCP stretch attempted if time (risk window W8–W10)          | 7/31/26 |
| Write-up complete; deliverables locked                                          | 8/7/26  |
| Final submission & project presentation; repo freeze + v1.0 tag (course)        | 8/12/26 |

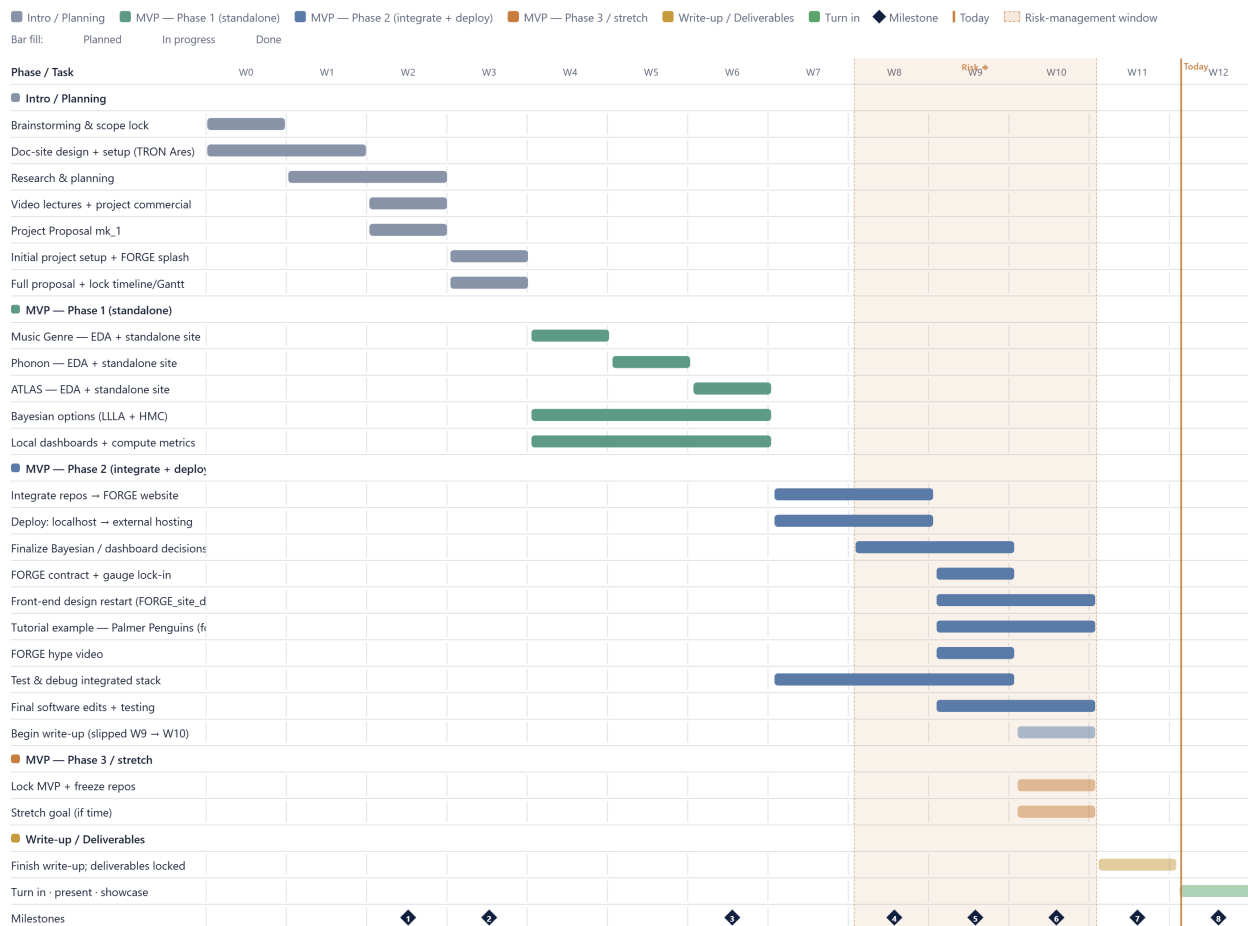
**Table 10:** Milestone Schedule (Summer 2026)

The Gantt chart in Figure 1 mirrors this schedule. Its source of truth is two hand-edited files in the doc-

umentation repository: `data/gantt.json` drives the bars (per-task phase, start/end week, and status) and encodes the shaded risk-management window via a `windows` entry, while `data/weekly_todos.json` holds each week's progress report — three lists per week (*planned*, *refactored*, *accomplished*) keyed to weeks W0–W12 — that populate the interactive Gantt's clickable week-detail panel on the project site and double as the weekly status update. A live version (clickable per-week detail, downloadable PNG) is maintained at the documentation site linked below.

FORGE — Project Gantt snapshot Aug 10, 2026

Thirteen-week project schedule (single contributor). Week 0 anchor: 2026-05-18. Bars span the weeks each task is active; diamonds mark milestones; the vertical line marks today.



#### Milestones

1. Proposal mk\_1 submitted — Week 2
2. Full proposal + timeline/Gantt locked — Week 3
3. All three MVP-P1 sites functional (guaranteed floor) — Week 6
4. Integrated FORGE site deployed (Phase 2) — Week 8
5. Front end restarted (select deck + splash); contract + gauges locked; write-up start → W10 — Week 9
6. MVP + repos locked — Week 10
7. Write-up complete; deliverables locked — Week 11
8. Final hand-in / showcase — Week 12

Last updated: 2026-07-29 | Source: `data/gantt.json` | Update workflow: edit JSON, commit, push.

**Figure 1: FORGE project Gantt (snapshot).** Bars span the weeks each task is active and are colored by phase (Intro/Planning, MVP Phase 1–3, Write-up/Deliverables); fill density encodes status (planned / in progress / done); the shaded band marks the risk-management window (W8–W10) and the vertical line marks “today”. Live version: [n-herling-mk1.github.io/INFO\\_698\\_documentation #gantt](https://n-herling-mk1.github.io/INFO_698_documentation #gantt).

## 5. Final Project — Final Stats

Prepared by Nathan Herling

Sections 1–4 above are the approved proposal, carried forward unchanged so that commitments and outcomes sit in one document. Everything from here to Section 8 is new.

**A note on the glossary.** Sections 5–8 carry three vocabularies at once — Bayesian inference, and then the particular language of a condensed-matter spectrum, a music benchmark and a displaced-vertex search. Every term, symbol, metric and equation used anywhere in this document is defined in the glossary at Section 13, grouped so it can be used as a reference rather than read: domain terms, instrument terms, metrics and tests, the three numbered equations, and a symbol table. A reader who meets an unfamiliar term should expect to find it there rather than hunt back through the text for its first use. This section closes the loop: §5a marks the deliverables met and unmet, §5b records how the instrument’s numerical contract evolved under development, and §5c gives the insights the build produced. Sections 6–8 then give the instrument, the measurements, and what they support.

### 5a Deliverables — met and unmet

Table 11 takes the feature list of Section 3 (Table 5) as it was written and marks each line against what was delivered. Table 12 does the same for the three project-level goals.

| #                     | Feature (as proposed)            | Status  | As delivered                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|----------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MVP — phase 1</b>  |                                  |         |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| F1                    | Preloaded multi-domain baselines | YES     | One documented baseline preloaded per domain, drawn from ten models built, all loading through one common interface, with a click-through tutorial alongside them.                                                                                                                                                                                                                                                                        |
| F2                    | Deterministic ↔ Bayesian toggle  | YES     | Delivered. The point estimate is the $\tau \rightarrow \infty$ limit of the same axis, so the two views are one continuum rather than two modes.                                                                                                                                                                                                                                                                                          |
| F3                    | Real-time knob panel             | CAVEAT  | Retired in favour of <i>solving</i> for the knob. $\tau^*$ is set by MacKay’s evidence criterion (MacKay, 1992; glossary, §13) — the value of $\tau$ that maximizes the marginal likelihood of the training data, found by fixed-point iteration and needing no held-out labels — so the value a user would have hunted for is computed and shown. The likelihood knob became an offline head sweep; the architecture knob was not built. |
| F4                    | Posterior-geometry visualization | YES     | Posterior bars with MAP ticks, GGN eigenspectrum, reliability and PIT diagnostics, all in-browser.                                                                                                                                                                                                                                                                                                                                        |
| F5                    | Resource-logging layer           | RETIRED | Not built, and not needed. F5 existed to size the metered cloud tier of F6 — a rented box plus serverless GPU. The stack redesign removed that tier, so there is no bill to predict and no autoscaling to inform.                                                                                                                                                                                                                         |
| <b>MVP — deployed</b> |                                  |         |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| F6                    | Cloud deployment                 | CAVEAT  | Deployed and public-live, but not on a cloud host: the rented tier was removed and the stack is tunnelled off a home box at \$0/month.                                                                                                                                                                                                                                                                                                    |
| <b>Stretch — MCP</b>  |                                  |         |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| S1                    | MCP server                       | NO      | Descoped. Declared optional in the proposal and not attempted; the risk window was spent on the metric lock-in instead.                                                                                                                                                                                                                                                                                                                   |

**Table 11:** Feature deliverables of Table 5, marked against what was delivered.

| Goal (as proposed)                                                                   | Met | On what evidence                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deploy inside a hobby-scale hosting envelope, ~\$25/month (Table 7)                  | YES | \$0/month. Verified public-live from a second machine; only recurring cost is domain registration.                                                                                                                                                                          |
| Turn a point estimate into a full Bayesian posterior — LLLA fast path, HMC reference | YES | Delivered on all four models at the scope §3b sets — a Bayesian <i>head</i> : a Gaussian posterior over the last-layer weights (LLLA; glossary, §13), with $\tau^*$ set by evidence rather than tuned, plus the exact functionals and per-feature rows that follow from it. |
| Generalize across heterogeneous domains, not one model                               | YES | Three deliberately unlike reproductions: CNN classifier, E(3)-equivariant GNN regressor, HEP binary classifier.                                                                                                                                                             |
| Reproduce each base model to its published or historical reference                   | YES | Phonon 0.717 vs Chen's 0.70; genre reproduces its native prediction exactly; ATLAS NN1 0.8785 vs ~0.86, NN2 0.9697 vs ~0.945.                                                                                                                                               |

**Table 12:** Project-level goals as proposed, and the verdict against each.

Three of these need a sentence more than a table cell allows.

**One model needed work before a posterior was possible; two did not.** Genre and ATLAS already ended in a linear layer — one feeding a softmax, the other a sigmoid — exactly the map a Laplace approximation needs, so the posterior was placed on the network as trained. That restriction is not a shortcut for speed: Kristiadi *et al.* (2020) show a last-layer Gaussian removes the asymptotic overconfidence of a ReLU MAP network, and Daxberger *et al.* (2021) find it competitive with ensembles at far lower cost.

The phonon network has no such layer. Its final operation is an equivariant convolution contracting hidden features against edge spherical harmonics, with no weight matrix between those features and the prediction — and LLLA needs one, since it places a Gaussian over the weights of a final linear map and reads the predictive variance off it. The fix was a *readout head*: a small linear layer trained on the frozen network, taking a fixed feature vector and reproducing its output — manufacturing the map the method requires without disturbing the model it measures. Here that vector is the network's own 51-bin prediction, so nothing upstream was retrained. It pays a second dividend: the head is linear-Gaussian, so its posterior is exactly the object LLLA computes, and agreement with an HMC reference (Neal, 2011, in the NUTS form of Hoffman and Gelman, 2014) is guaranteed by construction.

**Generalization carries a different cost on each model.** On phonon the guarantee has a flip side — a check that cannot disagree cannot inform, so FF2 reads nothing there, accepted deliberately when that head was chosen. On ATLAS the constraint is dimensional: a 129-dimensional last layer is a thin channel for epistemic signal, and its per-feature rows are latent activations rather than named physics inputs, so FF4 reports which *coordinate* carries the doubt, not which *variable*. On genre it is the benchmark rather than the mathematics: the reference is the author's own earlier coursework, so genre shows portability to a third architecture without an independent number to test against.

**F3 was retired, and the reader should register it as a deviation.** The knob answered a question — *what should  $\tau$  be?* — that the instrument answers for itself, and across iterations it kept resolving into a cluttered surface: every control moved a value toward the optimum, which is what a user wants anyway. Once focus settled on the variance readouts of §§6.1–6.2, the decision followed: compute  $\tau^*$  and hand the user the answer, not the search. Appendix E backs it — the evidence optimum sits on the accuracy

plateau, so the computed value is what a search would reach. The cost is H3, the proposal’s keystone hypothesis: steerable geometry is now evidenced by a measured  $\tau$  sweep, not by a user turning a dial. F5 died with the metered cloud tier it existed to size (§5c); S1 was never attempted.

## 5b Project Evolution — the FORGE four

The *FORGE four* — FF1–FF4 — are the four numerical capabilities the dashboard is built on. They are not a proposal commitment; Sections 1–4 name no such set. They emerged during the build and were re-cut twice.

They began as four separate objects descended from the proposal’s two-path architecture and hypotheses H1–H3: a fast path, a reference sampler, a curvature plot, a per-feature marginal. The first re-cut separated what a user *reads* from what is *computed* to feed it. The second, at the variance lock of 5 August, mattered more: fixing the base measure as variance made all four operations on a single budget — split, verify, source, locate. That lock is argued in §6.1, the shared budget in §6.2.

|                      | Was                                                                                                         | Is                                                                                                                                                                                                                | Metric it reports                                                                                                                                                                                                                            |
|----------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FF1</b><br>split  | <ul style="list-style-type: none"> <li>the LLLA fast path</li> <li>a stand-in for HMC</li> </ul>            | <ul style="list-style-type: none"> <li>the panel that answers <i>how much of this doubt is reducible at all?</i></li> <li>posterior bars, decomposition, reliability diagram (Guo <i>et al.</i>, 2017)</li> </ul> | <ul style="list-style-type: none"> <li><math>E_k = \phi^\top \Sigma_{(k,k)} \phi</math> against <math>A_k</math></li> <li><math>\text{EAR}_k = 10 \log_{10}(E_k/A_k)</math> dB</li> <li>reducibility <math>E_k/(E_k + A_k)</math></li> </ul> |
| <b>FF2</b><br>verify | <ul style="list-style-type: none"> <li>the HMC gold standard</li> </ul>                                     | <ul style="list-style-type: none"> <li>a run-gated panel answering <i>is the fast split honest, and which way does it lie?</i></li> <li>diagnostic table and trace plots</li> </ul>                               | <ul style="list-style-type: none"> <li><math>\varepsilon_k = (E_k^{\text{LLLA}} - E_k^{\text{HMC}})/E_k^{\text{HMC}}</math>, signed</li> <li>fidelity = <math>100(1 -  \varepsilon_k )</math> %</li> </ul>                                   |
| <b>FF3</b><br>source | <ul style="list-style-type: none"> <li>an eigenspectrum plot</li> </ul>                                     | <ul style="list-style-type: none"> <li>the panel that answers <i>does the reducible part come from data, or from leaked prior?</i></li> <li>cumulative curvature, bulk vs. outlier scatter</li> </ul>             | <ul style="list-style-type: none"> <li><math>c_i = (\phi \cdot v_i)^2 / (\lambda_i + \tau)</math></li> <li>prior-leak = <math>\sum_{\lambda &lt; \tau} c_i / \sum_i c_i</math></li> </ul>                                                    |
| <b>FF4</b><br>locate | <ul style="list-style-type: none"> <li>per-feature posteriors</li> <li>blocked on two candidates</li> </ul> | <ul style="list-style-type: none"> <li>the panel that answers <i>which inputs carry the reducible doubt, and how much each?</i></li> <li>feature-marginal table, correlation heat map</li> </ul>                  | <ul style="list-style-type: none"> <li><math>a_k = \phi_k(\Sigma \phi)_k</math>, with <math>\sum_k a_k = E</math> exactly</li> <li>concentration <math>100(1 - \text{PR})</math></li> </ul>                                                  |

**Symbols.**  $k$  indexes the resolving axis — the subdivisions the model’s output already has (class, energy bin, signal region); see Table 15.  $\phi$  features entering the readout head,  $\phi_k$  its  $k$ -th component.  $\Sigma$  posterior covariance over the last-layer weights,  $\Sigma_{(k,k)}$  its  $k$ -th diagonal block.  $\tau$  prior scale (the regularization set by evidence).  $\lambda_i, v_i$  eigenvalue and eigenvector of the Gauss–Newton curvature.  $A_k$  aleatoric (irreducible) variance;  $E_k$  epistemic (reducible) variance;  $E = \sum_k E_k$ .  $c_i$  the share of  $E$  carried by eigendirection  $i$ .  $\varepsilon_k$  signed relative error of the fast split against the sampler.  $a_k$  the share of  $E$  allocated to feature  $k$ . PR participation ratio, the spread of that allocation across features.

**Table 13:** The *FORGE four*: what each dashboard channel was first conceived as, what it became, and the number it now reports. Every metric named in the third column is defined in the glossary, §13.

## 5c Project Insights

Three things came out of the build that the proposal did not anticipate.

**ATLAS.** DNCn\_QN (glossary, §13), a closure term developed for this analysis and added to the double-DisCo loss of Kasieczka *et al.* (2021), bought threefold better ABCD background closure at fixed



distance correlation on endcap (§7.4).

**Phonon.** A ridge-regularized readout fitted on the frozen trunk beat the published baseline outright — frac@10% (glossary, §13) 0.717 against the 0.70 of Chen *et al.* (2021), and above our own reproduction at 0.711 (Appendix E).

**Deployment.** A static front end on GitHub Pages, with the live backends running in Docker on a home machine behind a tunnel, serves the whole observatory at \$0/month — no rented tier at all (§5a).

## 6. Research Component

*Prepared by Nathan Herling*

### 6.1 The base measure: variance, not entropy

The proposal committed to separating what a model does not know from what cannot be known, without committing to the arithmetic that separates them. That arithmetic was locked on 5 August 2026, and the lock is this project’s central methodological claim: *FORGE’s base measure is variance, under the law of total variance, not entropy or mutual information.*

$$\underbrace{\text{Var}(y_k | x)}_{\text{total}} = \underbrace{\mathbb{E}_\theta[\text{Var}(y_k | x, \theta)]}_{A_k \text{ aleatoric, irreducible}} + \underbrace{\text{Var}_\theta(\mathbb{E}[y_k | x, \theta])}_{E_k \text{ epistemic, reducible}} \quad (1)$$

Equation (1) is not ours. The identity is the law of total variance, classical and unconditional; conditioning it on the model parameters  $\theta$  to separate aleatoric from epistemic uncertainty is standard in Bayesian deep learning, operationalized for deep networks by Kendall and Gal (2017). Ours is the decision to adopt it as FORGE’s *base measure* in place of the entropy and mutual-information forms used until 5 August, and to report it *per k* rather than as a scalar. Each model already returns a structured answer rather than a single number [genre  $\rightarrow$  10 class scores; phonon  $\rightarrow$  51 energy bins; ATLAS  $\rightarrow$  signal regions; Table 15], and  $k$  indexes that structure: for phonon it is the energy bin, so Equation (1) returns a separate reading at each of the 51 spectrum bins instead of one number for the material (§6.3).

Four reasons, each of which cost something to establish. (i) *It works on a plain regressor.* Entropy is defined over a probability distribution, and a network trained under mean-squared error never declares one — it emits a number per bin with no noise model, so there is no density to take the entropy of. Variance asks only for the spread of the predictions, available either way; on the phonon regression as originally trained, the entropy form is not awkward but uncomputable. (ii) *The number is dimensionless.* Differential entropy is unit-dependent — rescaling the DOS axis shifts it by a constant — while a ratio of variances is invariant. (iii) *The decibel becomes literal:* a ratio of variances *is* a signal-to-noise ratio, so the  $10\log_{10}$  convention is derived rather than borrowed. (iv) *Mutual information violates axioms* a doubt measure should satisfy — not maximal under complete ignorance, not monotone under mean-preserving spreads, not invariant under location shifts. The label-wise variance form has none of these.

The switch was not free. The July gauge calibration on the genre bundle was entropy-based and had to be discarded, no conversion existing between the scales; the gain at  $2\times$  data changes from 29.29 % to exactly 50 %, because epistemic *variance* falls as  $1/N$  while the old figure was the standard-deviation

rate; and the verification gauge changes from a total-variation distance, which has no natural form for a DOS regression, to a signed relative error. Two readouts sit on top of the split — the *epistemic-to-aleatoric ratio* (EAR) in decibels, and the reducible share of the budget:

$$\text{EAR}_k = 10 \log_{10} \left( \frac{E_k}{A_k} \right) \text{ dB} \quad \text{reducibility}_k = \frac{E_k}{E_k + A_k} \quad (2, 3)$$

Provenance differs between the two, and it is worth stating. Equation (3) is an ordinary variance share and claims nothing. Equation (2) is an adaptation: the  $10 \log_{10}$  decibel convention for a ratio of powers is standard signal-processing practice, and what is introduced here is its use as an uncertainty readout — which is legitimate rather than decorative precisely because, under Equation (1),  $E_k/A_k$  is a ratio of variances and therefore literally a signal-to-noise ratio. No prior use of the epistemic-to-aleatoric ratio in decibels was found; it is claimed as original in §6.5 on that basis. EAR aggregates as a *ratio of sums*, never a mean of ratios, which diverges wherever the aleatoric term approaches zero.

## 6.2 Four capabilities, one budget

Under this base measure the four numerical capabilities stop being four unrelated numbers. Symbols used below are defined in the glossary (§13). Each is one operation on the same variance budget: FF1 *splits* it, FF2 *verifies* the split, FF3 *sources* the reducible part, FF4 *locates* it.

|     | Role                   | Core quantity                                                                                                                                                                                                | Question answered                                             |
|-----|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| FF1 | Split (fast path)      | $\Sigma = (H_{\text{GGN}} + \tau I)^{-1}$ , Kronecker-factored after Ritter <i>et al.</i> (2018); $E_k = \phi^\top \Sigma_{(k,k)} \phi$ (regression),<br>$E_k = \nabla p_k^\top \Sigma \nabla p_k$ (softmax) | How much of this prediction's doubt is reducible at all?      |
| FF2 | Verify (gold standard) | $\epsilon_k = (E_k^{\text{LLA}} - E_k^{\text{HMC}}) / E_k^{\text{HMC}}$ , signed                                                                                                                             | Is the fast split honest, and which way does it lie?          |
| FF3 | Source (curvature)     | $E = \sum_i c_i$ , $c_i = (\phi \cdot v_i)^2 / (\lambda_i + \tau)$ ; prior-leak<br>$= \sum_{\lambda < \tau} c_i / \sum_i c_i$                                                                                | Does the reducible part come from data, or from leaked prior? |
| FF4 | Locate (allocation)    | $a_k = \phi_k(\Sigma \phi)_k = \frac{1}{2} \phi_k \partial E / \partial \phi_k$ , with $\sum_k a_k = E$ exactly (Euler)                                                                                      | Which inputs carry the reducible doubt, and how much each?    |

**Table 14:** The FORGE four as four operations on one variance budget: the role each plays, the quantity it computes, and the question a reader takes from it. Symbols are defined in the glossary, §13.

The dashboard's four user-facing gauges G1–G4 — a later artefact than the proposal, distinct from the capabilities feeding them — all changed value with the base measure. G1 (provenance) becomes the epistemic share  $E/(E + A)$ , replacing a mutual-information fraction on a scale it cannot convert to: the July genre calibration was refitted, not rescaled. G2 (posterior) becomes the gain at  $2\times$  data, exactly 50 %. G3 (attribution) becomes FF2's signed relative epistemic-variance error. G4 (genealogy) becomes  $100(1 - \text{PR})$  from the FF4 allocation. There is deliberately no fifth gauge averaging the others, and accuracy is not a gauge.

FF4's allocation is exact —  $E$  is degree-2 in  $\phi$ , so Euler's theorem makes the per-feature shares sum to the total with no residual — and must be plotted *signed*, since anti-correlated features contribute negatively and absolute values hide the most interesting rows. It also closed a build blocker: the marginal-block route and the existing sensitivity code are the same object up to a factor of  $\phi_k$ , one



matrix–vector product on the cached  $\Sigma$ .

The regularization scale  $\tau$  is not tuned on test data. MacKay’s evidence criterion (1992) sets it: the marginal likelihood  $p(\mathcal{D} \mid \tau)$ , obtained by integrating the weights out, is maximized by fixed-point iteration on  $\tau = \gamma(\tau) / \|w\|^2$ , where  $\gamma(\tau) = \sum_i \lambda_i / (\lambda_i + \tau)$  counts the parameters the data constrains. Throughout,  $\tau$  is quoted relative to the curvature it regularizes,  $\tau = 10^k \lambda_{\max}$  with  $\lambda_{\max}$  the largest Gauss–Newton eigenvalue, so  $k$  is a decade exponent and  $k^*$  its value at the evidence optimum. On phonon  $k^* = -4.97$  — five decades below  $\lambda_{\max}$  — landing on the accuracy plateau: an optimum found with no held-out labels that agrees with the one found using them.

### 6.3 EAR is a field, not a scalar

The project’s most transferable idea is that the epistemic-to-aleatoric ratio of Equation (2) should be reported *along the subdivisions the model’s own output already has* [genre  $\rightarrow$  10 class scores; phonon  $\rightarrow$  51 energy bins; ATLAS  $\rightarrow$  signal regions], and only aggregated for a headline. That subdivision is what is meant throughout by the *resolving axis*: the dimension along which the problem already separates its answer into parts, in place before FORGE reads anything. Table 15 gives it for each model.

| Model       | Resolving axis      | Reads as                                                                      |
|-------------|---------------------|-------------------------------------------------------------------------------|
| Music genre | 10 classes          | which genres are model-limited vs. genuinely confusable                       |
| Phonon DOS  | energy bin $\omega$ | EAR( $\omega$ ) — which frequency ranges are data-limited vs. physics-limited |
| ATLAS LLP   | signal region       | whether an excess is ignorance or background                                  |

**Table 15:** The resolving axis of each model: the subdivision its own output already carries, and what an epistemic-to-aleatoric reading along it answers.

### 6.4 Novel domain research: ATLAS closure

The ATLAS strand is the one contribution here that is not a reproduction. The production search estimates background with the ABCD method, whose validity rests on the two discriminants factorizing. The double-DisCo scheme of Kasieczka *et al.* (2021) enforces this with a distance-correlation penalty,  $\text{BCE} + \lambda \cdot \text{dCor}^2$  — dCor being the distance correlation of Székely *et al.* (2007), zero only under genuine independence, applied as a trainable regularizer by Kasieczka and Shih (2020). To that we add DNCn\_QN, penalizing the prediction error at the cut in units of its own statistical noise, hinged to stay silent below one standard deviation. Two fixes were required before it worked: the hinge itself (the optimizer had been chasing a floor rather than the target) and drawing cut positions per batch so that training optimizes the grid the gate scores. Both changed together, so “the hinge fixed it” is *not* established — an ablation is open work (§8).

The design idea worth carrying forward is that the ABCD closure statistic can be planted as a functional, as the phonon head plants zero-point energy and heat capacity, so that  $\text{Var} \left[ N_A^{\text{pred}} / N_A^{\text{obs}} \right]$  falls out of the same covariance block and “how uncertain is the background estimate” becomes a first-class posterior reading rather than a separate study.

## 6.5 What is original here

Four constructions here are presented as original rather than as citations, and — the condition that matters — were each exercised on real data rather than merely defined. Three had no prior art that could be found; the fourth, (ii), is a new form of an idea that does, and says so.

(i) **EAR in decibels.** The epistemic-to-aleatoric ratio expressed as  $10\log_{10}(E/A)$ , which is a literal signal-to-noise ratio under the variance base measure rather than a borrowed analogy. Read throughout §7.

(ii) **The DNCn\_QN closure term.** A penalty on the ABCD prediction error at the cut, in units of its own statistical noise, hinged to stay silent below one standard deviation. The motivation precedes the measurement: distance correlation is global while ABCD closure lives in the sparse high–high corner, so a term aimed at the corner should do work the global penalty cannot. It does — at fixed distance correlation on endcap it buys threefold better closure (§7.4).

The originality claim here is narrower than for (i) and (iii), deliberately. Adding a non-closure term to a double-DisCo loss is *not* new: ABCDisCoTEC (CMS Collaboration, 2025) does exactly that, penalizing a metric proportional to  $(N_A N_D - N_B N_C)^2$ . What is introduced here is the *form* — normalizing the closure error by its own statistical noise rather than penalizing the raw discrepancy, and hinging it silent below one standard deviation, so the optimizer is not asked to close better than the data can resolve. That construction was arrived at independently and no prior use of it was found; the general idea was not ours. Two limits are also named rather than deferred: barrel is  $n = 1$  per cell and unproven, and the hinge and the randomized cut positions changed together, so which one fixed it is unestablished.

(iii) **The multinomial DOS likelihood** with an effective count from the energy grid — what gives that regression a declared likelihood, and therefore an aleatoric term, at all. Exercised as heads H4 and H5 (§7.2).

(iv) **The RRM reliability metric** of Appendix C (glossary, §13). Each configuration is scored by a penalty vector  $\mathbf{v} = [1 - A, s/s_{\max}, U/U_{\max}]$  — inaccuracy, fold instability, posterior spread — and ranked by  $\text{RRM} = 1 - \|\mathbf{v}\|$ . It is cohort-relative and therefore undefined for a lone model, which is the structural reason mk2 is a sweep rather than a single config. Selected on cross-validated columns before any test score was read, it picked a configuration that beats the faithful cfg\_14 reproduction on every axis (§7.3): CV mean  $0.686 \rightarrow 0.769$ , fold standard deviation  $0.0106 \rightarrow 0.0035$ , validation  $0.7467 \rightarrow 0.7933$ , test  $0.700 \rightarrow 0.760$ . The faithful reproduction fails its own acceptance gate; the RRM pick clears it.

## 7. Results

*Prepared by Nathan Herling*

### 7.1 The board

Ten models were built across the three domains: three on the first pass in each — genre mk1–mk3, phonon mk1–mk3, and the ATLAS rungs R1–R3 — and then a fourth phonon model, mk4, rebuilt after the variance lock so a FORGE-readable head could be fitted to it. That multiplicity is the research component; the genealogies are what §§7.2–7.4 and Appendix E draw on. FORGE proper is the product component and preloads *one* model per domain, the rest reported here rather than shipped

as defaults. Table 16 gives the reference each domain reproduces and the figure measured against it; complete architectures, objectives and training parameters for all ten builds are in Appendix D. Across all eight exported bundles the two structural identity residuals read  $\sim 3 \times 10^{-8}$  and  $\sim 1 \times 10^{-15}$ , and at a complete basis the heat-capacity functional is exact to machine precision: the arithmetic behind every number that follows closes on itself.

| Model / domain                                               | Reference                                                                                                      | Reference figure                   | Measured here                                                                                                       |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Phonon DOS</b> — E(3)-equivariant GNN, 51-bin spectrum    | Chen <i>et al.</i> (2021), <i>Direct Prediction of Phonon Density of States with Euclidean Neural Networks</i> | 0.70 frac@10%                      | trunk retrained from scratch: <b>0.711</b> ; MSE 0.0236, JS 0.0596, EMD-1 $25.68 \text{ cm}^{-1}$                   |
| <b>Music genre</b> — CNN on GTZAN spectrograms, $K = 10$     | BEARDOWN (Herling & Attili, 2025), CNN genre methodology after Sridhar (2024); target set in §3b               | $\geq 0.75$ val (cfg_14: 0.766)    | 5 models exported ( $d = 512/384/768/384/768$ ), best 0.855 val; bundle reproduces native prediction <b>exactly</b> |
| <b>ATLAS LLP</b> — two 128-128-1 MLPs, endcap, 27,995 events | Run-3 MSVtx displaced-vertex search (ANA-EXOT-2025-24); baseline per Schott (2024) as set in §3b               | NN1 $\sim 0.86$ , NN2 $\sim 0.945$ | <b>NN1 0.8785</b> (above reference for the first time), <b>NN2 0.9697</b>                                           |

**Table 16:** The three reproduction domains: the reference each was measured against, the figure that reference reports, and the figure achieved here. One model per domain is preloaded in the observatory; the full genealogies behind these numbers are reported in §§7.2–7.4 and Appendix E.

The published phonon checkpoint proved unusable — it is pickled against a pre-rewrite version of the equivariance library and its keys no longer match the current network — so the trunk was retrained from scratch.

## 7.2 Phonon — the head sweep, and the sharpest instrument result

mk4 exists to make phonon readable by the FORGE four metrics (FF1–FF4, §5b), which need a declared likelihood and a linear map for the posterior to sit on — neither of which mk1 has. Fitting a head that supplies both costs accuracy, and the operating point is chosen on a measured trade rather than assumed: the prior scale  $\tau$  moves predictive accuracy and exposable epistemic signal in opposite directions. The full sweep, the two named operating points and the 0.664-against-0.711 comparison are in Appendix E. Everything below is read at the operating point that sweep selects:  $k^* = -4.97$  (§6.2), rank 32,  $d = 1632$ .

Six likelihood heads were swept rather than one committed to up front. They differ in two things only — the basis the readout is written in, and the likelihood declared over it — so Table 17 isolates what each choice costs. Read the last column first: two heads fail the aleatoric referee outright, and no amount of accuracy or coverage rescues them.

| Head                                 | Likelihood                                  | frac@10% | PICP95       | PIT-KS       | A-invariance       |
|--------------------------------------|---------------------------------------------|----------|--------------|--------------|--------------------|
| H3 planted-band ( <b>ships</b> )     | Gaussian, planted basis, banded $\Lambda$   | 0.664    | 0.886        | <b>0.149</b> | PASS               |
| H2 planted-diag ( <i>reference</i> ) | Gaussian, planted basis, diagonal $\Lambda$ | 0.664    | 0.889        | 0.152        | PASS               |
| H6 ( <i>control</i> )                | Gaussian, PCA basis, diagonal               | 0.664    | 0.888        | 0.154        | PASS               |
| H4                                   | multinomial                                 | 0.355    | —            | —            | <b>FAIL 184.4%</b> |
| H5                                   | multinomial + representation floor          | 0.355    | —            | —            | PASS 0.2%          |
| H7                                   | learned $\sigma^2$                          | 0.599    | <b>0.951</b> | —            | <b>FAIL 165.6%</b> |

**Table 17:** The mk4 head sweep at the instrument operating point ( $k^* = -4.97$ , rank 32,  $d = 1632$ ). H3 ships; H2 is the diagonal reference and H6 the PCA control. A dash means the quantity is not defined for that likelihood.

**H7 is the result to remember.** It posts PICP95 = 0.951 — dead on nominal, the best-calibrated head on the board — and an EAR of  $-53$  dB saying nothing is reducible, and it is *wrong*: the A-invariance referee (glossary, §13) catches it at 183.9% drift. Calibration does not detect bias contamination, and neither does EAR, which the contamination silently inflates on the aleatoric side. Only the invariance test catches it. H4 fails the same way at 165.9%, while H5 — the same head with a representation floor — passes at 0.2%, a sharper argument for the floor than anticipated.

**H3 ships.** The three gated Gaussian heads are indistinguishable on accuracy — all read frac@10% 0.664, with the gated-subset spread (0.559–0.586) inside binomial noise on 152 materials — so the decision rests on calibration and on what the covariance represents. H3 posts the board’s best PIT-KS (0.149 against 0.152 and 0.154) for 0.003 of PICP95, and its banded  $\Lambda$  alone models correlation between neighbouring energy bins — a property the target has, since a phonon spectrum does not vary independently bin to bin. H2, the diagonal reference, is the control showing what banding buys; both use the planted basis, so both carry exact zero-point-energy and heat-capacity posteriors and named FF4 rows.

Five hypotheses were tested and refuted: a peak-normalization penalty (+0.000; the target is scale-invariant), clipping negative predictions ( $-0.059$ ; a quarter of the cells are negative, compensating for high-frequency leakage), a  $\sigma$ -weighted loss, a linear-readout trunk (0.605 — a 0.106 cost for a real linear last layer), and an invariants surrogate (0.197–0.224, the network having no linear last layer to begin with, §5a). Three “properly architected” trunks all lost to the trained model’s own output as the feature map.

### 7.3 Genre — what reliability selection buys

The genre genealogy asks a different question from phonon’s. All three share the architecture; what varies is first *how a configuration is chosen*, then *how much data it is given*. Numbers below are read from each run’s `metrics.json`.

| Model                    | CV mean $\pm$ s.d. | Val           | Test          | What it establishes                                                                                                            |
|--------------------------|--------------------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| mk1 <i>beardown</i>      | $0.686 \pm 0.011$  | 0.7467        | 0.700         | The faithful cfg_14 reproduction — and it <i>fails</i> its own acceptance gate (0.7467 against a 0.75 floor).                  |
| mk2 <i>beardown_rrm</i>  | $0.769 \pm 0.004$  | <b>0.7933</b> | 0.760         | Same pipeline, selected by RRM rather than accuracy. Passes.                                                                   |
| mk3 <i>beardown_3sec</i> | $0.778 \pm 0.009$  | 0.7764        | <b>0.7887</b> | The data-limited control: $\sim 10\times$ the rows, split on track ids. Passes, and posts the best test accuracy on the board. |

**Table 18:** Genre genealogy. Accuracy is the headline, but the fold spread and the epistemic term are what the genealogy was built to move.

Three readings follow, and only the first is about accuracy.

**The faithful reproduction is the weakest model.** mk1 — the reproduction of Nathan Herling’s final project for [INFO 510, Bayesian Modeling and Inference](#) — reproduces cfg\_14 as specified and lands at 0.7467 validation, below the 0.75 acceptance floor set in §3b. That is reported rather than quietly re-run: the target came from the original project, and missing it by 0.003 is a fact about the reproduction, not a defect to be tuned away.

**Reliability selection wins outright.** mk2 differs from mk1 only in the selection rule — RRM instead of validation accuracy — and it is better on every axis measured: CV mean  $0.686 \rightarrow 0.769$ , validation  $0.7467 \rightarrow 0.7933$ , test  $0.700 \rightarrow 0.760$ , and fold standard deviation  $0.011 \rightarrow 0.004$ , a factor of three tighter. Selecting for reliability did not cost accuracy to buy stability; it produced more of both. The caveat is on the third axis rather than the result: across this cohort the posterior-spread term  $U$  is near-constant, so the ranking is carried by accuracy and fold stability and the epistemic axis had nothing to separate — which is also why the locked equal-weight setting ranks configurations identically to a plain norm (Appendix C). RRM is used for selection only, never for reporting.

**More data raises accuracy and shrinks the epistemic term.** mk3 fixes architecture, sweep space and selection and gives ten times the rows: test accuracy rises to 0.7887, the highest here, and the selector’s  $U$  falls from 0.027 to 0.0092 — the epistemic axis does not revive with more data, it recedes. One caveat the metrics do not carry: in informal use, the data-poorer models gave richer cross-genre predictions for songs that sit between the ten labels, where mk3 commits harder to one.

#### 7.4 ATLAS — closure, and the referee

Three rungs were trained on 27,995 endcap events: vanilla BCE; plus the double-DisCo penalty of Kasieczka *et al.* (2021); plus the DNCn\_QN closure term. In a single-seed sweep over  $\lambda$  the decorrelation penalty does real work — endcap non-closure falls from 223% to 4.8% at  $\lambda = 5$ , a  $46\times$  improvement for 0.002 NN1 and 0.009 NN2 AUC — then stalls against the two-sigma gate: dCor keeps falling, closure stops following. A replicated  $3 \times 4 \times 3$  factorial ( $\lambda \in \{2, 3, 5\}$ ,  $\mu \in \{0, 1, 2, 5\}$ , three seeds) resolves why. Its cells are the numbers below; they are seed-medians rather than single runs, which is why  $\lambda = 5$ ,  $\mu = 0$  reads 6.8% there against the sweep’s 4.8%.

| Endcap operating point                                        | dCor        | $ nc _{\text{med}}$                   | within $2\sigma$ | NN1 AUC      |
|---------------------------------------------------------------|-------------|---------------------------------------|------------------|--------------|
| $\lambda = 5, \mu = 0$ (DisCo only)                           | 0.0431      | $0.0675 \pm 0.0179$                   | 38.9 %           | 0.8775       |
| $\lambda = 5, \mu = 5$ ( <b>production</b> )                  | 0.0430      | <b><math>0.0216 \pm 0.0069</math></b> | <b>88.9 %</b>    | $\sim 0.867$ |
| $\lambda = 2, \mu = 1$ (best closure, <i>fails</i> dCor gate) | 0.059–0.064 | $0.0213 \pm 0.003$                    | 88 %             | 0.8692       |

**Table 19:** Three cells of the endcap factorial, seed-medians over three seeds. The first two share a distance correlation and differ only in  $\mu$ , which isolates what the closure term buys; the third has the best closure on the board and is unusable because it fails the dCor gate.

At the same distance correlation, adding the closure term gives three times better closure. The mechanism is that the two penalties are not doing the same job:  $\mu$  moves the closure statistic by 20.0 standard errors while moving dCor by only 2.8, *in the wrong direction*;  $\lambda$  moves dCor by 9.1 standard errors. Distance correlation is a global statistic dominated by the bulk where the events are, while ABCD closure depends on the sparse high–high corner. One can drive global dependence close to zero and still be badly off in the corner. The cost is 1.4 % relative NN1 AUC (28.8 standard errors — small, but real and measured, not zero).

Every ATLAS result claimed here is an endcap result, and the endcap is where the evidence supports one: three seeds per cell, gated on dCor, reported as measured. Barrel is not claimed. Its mid-point closure (0.057) is comparable, but with 2,280 background events against 24,850 the error bars are wide enough to swallow an error the endcap flags — a barrel pass is weaker evidence than an endcap fail, not stronger — and the barrel sweep was  $n = 1$  per cell. It is carried into §8 as work, not a finding.

## 7.5 The deployed observatory

Four public repositories carry the project, deliberately separated so that development never happens where deployment does. [INFO\\_698\\_experiments](#) is the research tier: every model in Appendix D is trained there, and nothing user-facing is served from it. [FORGE\\_site\\_dev](#) is the design tier, added at the Stage-C front-end restart — interface, flow and motion with all data stubbed, so layout could be iterated without touching a model. [forge-observatory](#) is the deploy tier and the origin behind [forge-observatory.com](#): an app matures through its mk cycle in the experiments repository, and only a locked version is copied into apps / here, where Docker Compose serves it. And [INFO\\_698\\_documentation](#) carries the site this is published from and serves the apex domain over GitHub Pages, while the sub-domains resolve through a Cloudflare tunnel to the home box. Its own front end is live and browsable at [n-herling-mk1.github.io/INFO\\_698\\_documentation](#), which is where the Gantt of Figure 1, the Literature Search panel behind §2b and the weekly progress reports are read. Appendix F gives the deploy repository’s structure and how a request reaches a model.



**Figure 2:** The splash page. A single strike ignites the lockup and drops the visitor into the model deck; the site is served at zero hosting cost from a tunnelled home box.



**Figure 3:** The navigation deck. Four model plates orbit a holographic projection; the left rail carries a cathode-terminal panel — showing the control legend here — above the model selector, and the beam locks onto whichever plate is under it.





**Figure 4:** The FORGE dashboard. Left rail selects  $\tau$  and the four capabilities FF1–FF4; the right pane carries the epistemic reading, an explanation of how that metric fits the whole, and the graph area.

## 8. Conclusion and Future Work

*Prepared by Nathan Herling*

FORGE set out to build an observatory in which the relationship between the data, the error it carries, and the architecture trained on it is built into a dashboard a user can actually read. That resolved into two missions: to build a Functional-posterior Observatory for Refinement and Geometric Exploration that performs as promised, and to give it an interface that encourages a user to engage with it. As a data-science capstone it succeeds at both, fusing the mathematics and statistics of the degree with its data analysis and visualization component.

**The research delivered on four fronts.** The models: each was rebuilt and driven to its reference before anything Bayesian was attached — phonon 0.711 against Chen’s 0.70, ATLAS NN1 0.8785 and NN2 0.9697 against  $\sim 0.86$  and  $\sim 0.945$ , genre reproducing `cfg_14`. The mathematics: a variance-based uncertainty budget (§6.5) that ported to a spectrum regressor, a ten-class audio classifier and a HEP binary classifier with only the modifications each architecture named, reporting as a *field* along each domain’s own axis rather than as a scalar — Table 15 is the transferable object. The new metrics: DNCn\_QN bought threefold better closure at fixed distance correlation; RRM chose a configuration beating the faithful reproduction on every axis; and the invariance referee caught a planted control, H7, posting perfect coverage while reporting nothing reducible. Coverage alone is not evidence of a trustworthy posterior. And, least expected, the modifications themselves: what makes a model FORGE-readable — a declared likelihood, a linear map for the posterior — is a research direction of its own, and the one the next release is built on.

**Where this sits.** FORGE builds on surveyed methods. Deep ensembles (Lakshminarayanan *et al.*, 2017) and probabilistic backpropagation (Hernández-Lobato and Adams, 2015) are the non-Laplace baselines a posterior tool is judged against; Izmailov *et al.* (2021) and Wilson and Izmailov (2020) establish what true posteriors look like and why marginalization is the essential move; calibration eval-



uation descends from Pakdaman Naeini *et al.* (2015) and degrades under shift (Ovadia *et al.*, 2019); the surveys (Gawlikowski *et al.*, 2023; Papamarkou *et al.*, 2024) frame why it matters at scale. The gap FORGE occupies lies between that work and the visualization literature, which probes point predictions and static displays (Hullman *et al.*, 2015; Hohman *et al.*, 2019; Wexler *et al.*, 2020) but not posterior structure.

**Product: the interface is the contribution the mathematics cannot make.** A splash, a model deck, then one dashboard whose rails do not change between models — so a phonon spectrum and a displaced-vertex search read in one vocabulary. It does not ask a user to hunt for the prior scale; it solves for it and shows the answer. Four channels become four readings a practitioner can act on: how much doubt is reducible, whether the fast estimate is honest, whether it came from data or prior, and which inputs carry it.

**That fusion is the result.** The mathematics is only useful if it can be read; the interface is only worth building over numbers that survive scrutiny. FORGE shows that data-science method and interface design are one problem, not two.

**Future work.** The clearest direction is to need less from the model. Phonon needed a readout head because its final operation leaves no weight matrix for a posterior to sit on; a method reading curvature directly from that operation would remove the modification and the accuracy cost Appendix E measures. That is the `mk_2` problem. Second, more variance metrics: the law of total variance admits readouts beyond EAR, reducibility and the four built here, each another question asked of the same cached  $\Sigma$  at no extra compute. Information-theoretic readouts are worth revisiting as additional channels, not as the base measure, and the noise floor deserves its own study — the representation floor is declared, not measured. Third, more models: three domains establish portability, and the architectures that resist are the more interesting result, each naming a new requirement. Fourth, the product goal: general ingestion — a researcher uploads their own model, the contract decides which channels apply, and any channel that cannot speak is declared unavailable with a reason. And finally, the barrel closure test needs replicating: `DNCn_QN` is proved on endcap alone, and the barrel is where the technique still wants vetting.

## 9. References

*Prepared by Nathan Herling*

Entries below are the project bibliography, maintained on the documentation site’s Literature panel and reproduced here so the report is self-contained. Every entry was confirmed against arXiv, the official proceedings, or the publisher of record; where proceedings page numbers could not be independently confirmed they are omitted rather than guessed.

### **Inference — variational, MC-dropout, ensembles**

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Gal, Y. & Ghahramani, Z. (2016). *Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning*. ICML 2016, PMLR 48:1050–1059. arXiv:1506.02142.

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Ritter, H., Botev, A. & Barber, D. (2018). *A Scalable Laplace Approximation for Neural Networks*. ICLR 2018. OpenReview:Skdvd2xAZ.

Kristiadi, A., Hein, M. & Hennig, P. (2020). *Being Bayesian, Even Just a Bit, Fixes Overconfidence in ReLU Networks*. ICML 2020, PMLR 119:5436–5446. arXiv:2002.10118.

Daxberger, E., Kristiadi, A., Immer, A., Eschenhagen, R., Bauer, M. & Hennig, P. (2021). *Laplace Redux — Effortless Bayesian Deep Learning*. NeurIPS 2021 (vol. 34), 20089–20103. arXiv:2106.14806.

### **Sampling and posterior geometry**

Neal, R.M. (2011). *MCMC using Hamiltonian Dynamics*. Ch. 5 in Handbook of Markov Chain Monte Carlo (Brooks, Gelman, Jones & Meng, eds.), Chapman & Hall/CRC, 113–162. DOI:10.1201/b10905. arXiv:1206.1901.

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Izmailov, P., Vikram, S., Hoffman, M.D. & Wilson, A.G. (2021). *What Are Bayesian Neural Network Posteriors Really Like?* ICML 2021, PMLR vol. 139. arXiv:2104.14421.

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### **Calibration and predictive-uncertainty evaluation**

Pakdaman Naeini, M., Cooper, G.F. & Hauskrecht, M. (2015). *Obtaining Well Calibrated Probabilities Using Bayesian Binning*. AAAI 2015 (29th), 2901–2907. DOI:10.1609/aaai.v29i1.9602.

Guo, C., Pleiss, G., Sun, Y. & Weinberger, K.Q. (2017). *On Calibration of Modern Neural Networks*. ICML 2017, PMLR 70:1321–1330. arXiv:1706.04599.

Kendall, A. & Gal, Y. (2017). *What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?* NeurIPS 2017 (vol. 30). arXiv:1703.04977.

Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J.V., Lakshminarayanan, B. & Snoek, J. (2019). *Can You Trust Your Model’s Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift*. NeurIPS

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Hohman, F., Kahng, M., Pienta, R. & Chau, D.H. (2019). *Visual Analytics in Deep Learning: An Interrogative Survey for the Next Frontiers*. IEEE TVCG 25(8):2674–2693. DOI:10.1109/TVCG.2018.2843369. arXiv:1801.06889.

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Nussbaumer, N., Böck, M. & Cito, J. (2026). *Online and Interactive Bayesian Inference Debugging*. ICSE 2026 (IEEE/ACM 48th International Conference on Software Engineering), 12 pp. DOI:10.1145/3744916.3773215. arXiv:2510.26579. The InferLog Holmes debugger; closest prior art to FORGE’s online, interactive ambition (Table 4).

### Surveys and state-of-the-art positioning

Gawlikowski, J., Njieutcheu Tassi, C.R., Ali, M., Lee, J., Humt, M., Feng, J., Kruspe, A., Triebel, R., Jung, P., Roscher, R., Shahzad, M., Yang, W., Bamler, R. & Zhu, X.X. (2023). *A Survey of Uncertainty in Deep Neural Networks*. Artificial Intelligence Review 56:1513–1589. DOI:10.1007/s10462-023-10562-9. arXiv:2107.03342.

Papamarkou, T., Skoularidou, M., Palla, K., Aitchison, L., Arbel, J., Dunson, D., Filippone, M., Fortuin, V., Hennig, P., Hernández-Lobato, J.M., Hubin, A., Immer, A., Karaletsos, T., Khan, M.E., Kristiadi, A., Li, Y., Mandt, S., Nemeth, C., Osborne, M.A., Rudner, T.G.J., Rügamer, D., Teh, Y.W., Welling, M., Wilson, A.G. & Zhang, R. (2024). *Position: Bayesian Deep Learning is Needed in the Age of Large-Scale AI*. ICML 2024, PMLR 235:39556–39586. arXiv:2402.00809.

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*tion with LHC data: ABCDisCoTEC*. arXiv:2506.08826. Preprint — adds a non-closure term to the double-DisCo loss; see §6.4.

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Petretto, G., Dwaraknath, S., Miranda, H.P.C., Winston, D., Giantomassi, M., van Setten, M.J., Gonze, X., Persson, K.A., Hautier, G. & Rignanese, G.-M. (2018). *High-throughput density-functional perturbation theory phonons for inorganic materials*. *Scientific Data* 5, 180065. DOI:10.1038/sdata.2018.65.

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Schott, M. (forthcoming). *Long-Lived Particle searches via displaced muon-spectrometer vertices*. Doctoral thesis. Full bibliographic record to be finalized in a later revision.

## 10. Ethics

*Prepared by Nathan Herling*

**ATLAS data note.** Whether the ATLAS/CERN detector data underlying the LLP reproduction may be released publicly is still being vetted under the collaboration's data-access and publication policy (with advisor input). If public release is not permitted, the fallback is to keep the data private and release only the model and methodology — which is sufficient for the FORGE demonstration, since the platform's contribution is the reliability-aware methodology rather than the raw detector inputs.

| #  | Question                                                                                                                                                      | Generally                              | Data Breach                            |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------|
| 1  | Could a user sell drugs or other illegal items on your platform?                                                                                              | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 2  | Could a user of your platform engage in sex trafficking?                                                                                                      | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 3  | Could a user sell class notes or cheat on their homework on your platform?                                                                                    | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 4  | Could a stalker use your project to find someone?                                                                                                             | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 5  | Could your app be used to spy on or track individuals?                                                                                                        | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 6  | Could your app/software access the camera or microphone and record things without users being aware?                                                          | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 7  | If someone uses your platform, could they be re-traumatized or have their mental health impacted in some way?                                                 | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 8  | Could your algorithm promote material that would traumatize or upset individuals?                                                                             | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 9  | Would your users be upset if the data you collect was given to someone else?                                                                                  | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 10 | Could a data leak potentially lead to identity theft?                                                                                                         | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 11 | If your site was hacked, would users potentially lose their job, spouse, or family?                                                                           | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 12 | Should there be an age limitation on your product?                                                                                                            | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 13 | Could someone use your product to find, contact, and potentially commit elder abuse?                                                                          | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 14 | If the data on your platform was breached, could it be used to blackmail the users?                                                                           | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 15 | Does the existence of your project imply that a particular racial group, gender, religion, or other protected category is inherently bad, gross, or unwanted? | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |
| 16 | Could your product be used to commit hate crimes against a specific group?                                                                                    | Y/ <input checked="" type="radio"/> /M | Y/ <input checked="" type="radio"/> /M |

| #  | Question                                                                                                                                   | Generally                               | Data Breach                             |
|----|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------|
| 17 | Does the primary content of your game or algorithm focus on something considered deeply unethical?                                         | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 18 | Does your game or software contain race, gender, or other stereotypes?                                                                     | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 19 | Could users of your app scam other individuals?                                                                                            | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 20 | Is your particular algorithm biased toward predicting correctly only for one race, gender, or other group?                                 | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 21 | Are the users of your project aware of how their data will be used?                                                                        | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 22 | What are the possible misinterpretations of your results?                                                                                  | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 23 | Does the use or purchase of your data potentially contribute to a dangerous group or regime?                                               | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 24 | Could your virtual reality environment cause injury to the user?                                                                           | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 25 | Are your study participants or game players aware that their data will be collected and used?                                              | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 26 | Does your game or app contain addictive design elements without benefit to the user?                                                       | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 27 | Does your survey contain an aspect of compulsion or unusually large incentive that would command users to take it even to their detriment? | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |
| 28 | Could your research outcomes harm an individual or entity?                                                                                 | Y/ <input checked="" type="radio"/> N/M | Y/ <input checked="" type="radio"/> N/M |

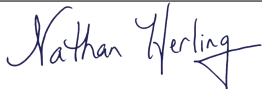
**Table 20: Data Ethics Chart**

## 11. Approvals

*Prepared by Nathan Herling*

The signatures below indicate an understanding of the purpose and content of this document by those signing it. By signing, you indicate that you have reviewed the work reported here against the commitments made in the approved proposal of 12 June 2026 — Sections 1–4, carried forward unchanged — and that the verdicts recorded in Section 5, the results in Section 7, and the open items carried into Section 8 are an accurate account of the FORGE capstone as delivered.

This document is self-contained and supersedes no part of the approved proposal: it extends it. The proposal's Statement of Work remains the governing specification, and every change since approval is tracked through the Revision History above. Deviations from the signed scope are stated where they occur and are summarized in Section 5a.

| Approver Name      | Title           | Signature                                                                          | Date     |
|--------------------|-----------------|------------------------------------------------------------------------------------|----------|
| Nathan Herling     | Project Lead    |  | 08/10/26 |
| Ken Johns, PhD     | Faculty Advisor |                                                                                    |          |
| Nitika Sharma, PhD | Instructor      |                                                                                    |          |

| Section                                                          | Author         | Word Count     |
|------------------------------------------------------------------|----------------|----------------|
| <b>Carried forward from the approved proposal</b>                |                |                |
| 1. Executive Summary (incl. 227 new)                             | Nathan Herling | 673            |
| 2 / 2b. User/Market & Literature Review                          | Nathan Herling | 1,269          |
| 3 / 3b. Product Features & Research Deliverables (incl. 457 new) | Nathan Herling | 2,301          |
| 4. Timeline & Gantt (incl. tables/figure)                        | Nathan Herling | 444            |
| <b>Written for this report</b>                                   |                |                |
| 5. Final Project — Final Stats                                   | Nathan Herling | 1,677          |
| 6. Research Component                                            | Nathan Herling | 1,906          |
| 7. Results                                                       | Nathan Herling | 2,000          |
| 8. Conclusion & Future Work                                      | Nathan Herling | 671            |
| <b>Reference and back matter</b>                                 |                |                |
| 9. References                                                    | Nathan Herling | 999            |
| 10 / 11. Ethics & Approvals                                      | Nathan Herling | 1,020          |
| 12. Appendix A–F (D, E, F new)                                   | Nathan Herling | 3,343          |
| 13. Glossary                                                     | Nathan Herling | 2,357          |
| <b>Graded body, Sections 1–8</b>                                 |                | <b>10,941</b>  |
| <b>Whole document</b>                                            |                | <b>~18,650</b> |

Counts exclude inline citations. This is a solo capstone: Nathan Herling is the sole author of every section, table, figure and appendix listed above, and the sole contributor to the four repositories linked in Section 7.5.

## 12. Appendix

*Prepared by Nathan Herling*

### A. Advisor Engagement

## Project Team Responsibilities

- The Project Manager will set up and facilitate a weekly call/meeting with the Faculty Advisor. The Project Team will provide weekly status updates including upcoming deliverables, critical issues, and any adjustments to the Project Plan.
- Documents will be provided to the Faculty Advisor with adequate time for review and signature, with a minimum review time of 3 days prior to the document due date.
- Design files will be provided to the Faculty Advisor as requested in an agreed format.
- Support requirements will be clearly requested with the dates required and adequate time for fulfillment.
- Modification requests to the Project Plan by the Faculty Advisor will be reviewed and agreed to within 1 week of the request.

## Faculty Advisor Responsibilities

- The Faculty Advisor will provide knowledge and expertise to help the group stretch their skills.
- The Faculty Advisor will participate in a weekly or bi-weekly call/meeting to review status, upcoming deliverables, priorities, issues, and progress to the agreed Project Plan.
- The Faculty Advisor will provide document review, feedback, and approval (or rejection, or approval with contingencies) with adequate time for the team to meet course due dates.
- The Faculty Advisor will provide feedback on support requirements, including guidance on design decisions, design files, test plans, test procedures, and test results.
- The Faculty Advisor shall provide technical advice and guidance, answering inquiries approximately 1 hour per week.
- Modifications to the Project Plan by the Project Team will be resolved and documented within 1 week of the request.
- Grade the finalized project using a skill-based rubric.
- Attend the project presentation.

## B. Ground Rules

As a team and as individual team members, we agree to:

- **Stay focused on our objectives and goals.** Each time the team meets, we will clearly define our objectives and desired outcomes at the beginning of the meeting, and politely remind team members if we are getting off track.
- **“Sidebar” issues that are relevant but not consistent with the immediate objectives.** Create a “sidebar” where off-topic but important matters can be listed and discussed later.
- **Listen when others are speaking.** We will listen and consider others’ input before adding our own comments.
- **All viewpoints will have an opportunity to be heard.** We will make an effort to get each team member’s viewpoint and ensure no one dominates the discussion.
- **Differences of opinion will be discussed respectfully.** We will identify areas of agreement before assessing disagreement, weigh the merits of different opinions, and agree on a process for choosing a direction. All team members will respect and follow the decision.
- **Look for the good points in new ideas.** We will explore the value in each idea as we assess and select our path forward.
- **Focus on the future, not the past.** We will use past experience to inform decisions but focus the discussion on future objectives; blame is counterproductive.
- **Agree upon specific action items and next steps.** At the end of each meeting we will summarize and agree on specific next steps, action items, and assignments.
- **Accountability.** Each member is responsible for their individual assignments and contribution to the team objectives. We will honor our responsibilities and not let our team members down.



## C. Exploratory Reliability-Aware Selection Metrics

**Source.** The code and the full BEARDOWN report — including Figures 6–7 and the metric tables of its Appendix B — are public: [BEARDOWN repository](#) and [BEARDOWN report \(PDF\)](#).

This appendix collects an author-developed, *exploratory* suite of selection metrics: proof-of-concept rather than established canon, but each resting on a stated hypothesis with initial empirical support. They are used only to *steer model selection during hyperparameter sweeps*, never to report results — Section 3b’s numbers use standard metrics (accuracy, precision/recall/F1, ROC–AUC, ECE) and stay comparable to published baselines, while the selection criterion is free to be more opinionated. The premise: accuracy alone is a poor sweep objective when the target is a *reliable* model, since a configuration can top validation accuracy while being unstable across folds or badly miscalibrated — exactly the failure mode FORGE exists to expose.

**Robust Ranking Metric (RRM) — the anchor.** For a configuration scored under  $k$ -fold cross-validation, form a three-axis penalty vector measuring inaccuracy, instability, and posterior uncertainty, and aggregate by Euclidean distance from the ideal  $\mathbf{v} = \mathbf{0}$ :

$$\mathbf{v} = \left( 1 - \bar{a}, \frac{\sigma_{\text{fold}}}{\sigma_{\text{max}}}, \frac{U_{\text{mean}}}{U_{\text{max}}} \right), \quad \text{RRM} = 1 - \|\mathbf{v}\|_2.$$

Here  $\bar{a}$  is the mean fold accuracy,  $\sigma_{\text{fold}}$  the standard deviation of the per-fold accuracies, and  $U_{\text{mean}} = \text{mean} \sqrt{\text{diag} \Sigma}$  the mean posterior standard deviation of the Bayesian head. Selection *maximizes* RRM; it reaches 1 only for a perfectly accurate, stable, zero-uncertainty model, so nothing can win on accuracy alone while unstable or diffuse. The stability and uncertainty axes are normalized by running sweep maxima ( $\sigma_{\text{max}}, U_{\text{max}}$ , floored at  $10^{-8}$ ) onto a common  $[0, 1]$  scale; the uncertainty axis is the same posterior signal FORGE visualizes, here turned into an objective. Locked and run in BEARDOWN over two 20-configuration, 3-fold sweeps (mk51/mk5m).

**Preliminary evidence — non-redundancy.** RRM earns its place only if it carries information the standard metrics do not. In BEARDOWN (Herling & Attili, 2025, Figs. 6–7) RRM correlated with accuracy at only  $r \approx 0.62$  ( $R^2 \approx 0.38$ , and partly by construction, since accuracy is one of its components), negatively with the uncertainty axes ( $r \approx -0.33, -0.37$ ), and was statistically indistinguishable from uncorrelated with an *external* diagnostic not built into it — off-diagonal confusion noise ( $r \approx 0.26, p \approx 0.26$ ). No single existing metric reproduces its ranking: several high-accuracy configurations scored low on RRM, and some lower-accuracy ones ranked higher. The honest reading is *non-redundant*, not orthogonal — RRM occupies a distinct direction in metric space rather than re-expressing accuracy. Because Pearson  $r$  captures only *linear* association while RRM is a nonlinear (norm) statistic, the planned next diagnostic is distance correlation (dCor, and partial-dCor controlling for the shared accuracy term), which measures dependence without the linearity assumption; the present correlations are exploratory, small- $n$  estimates.

**Open question — the weighting.** Collapsing three heterogeneous axes into one scalar requires a relative weight per axis,  $\mathbf{w} = (w_0, w_1, w_2)$ , and a fixed choice is both arbitrary and game-able. Because only *relative* emphasis affects the ranking, the weights are normalized to sum to one — i.e. constrained to the probability simplex  $\Delta^2 = \{\mathbf{w} : w_i \geq 0, \sum_i w_i = 1\}$ , the triangle whose three corners place all weight on a single axis and whose centroid weights the three equally:

$$\text{RRM}_{\mathbf{w}} = 1 - \sqrt{\sum_i w_i v_i^2}, \quad \mathbf{w} \in \Delta^2.$$

The locked metric is the equal-weight case — the centroid  $\mathbf{w} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$  — which ranks configurations identically to  $1 - \|\mathbf{v}\|_2$ . Rather than fix  $\mathbf{w}$ , the intended treatment *samples* it: each weighting is drawn from a Dirichlet distribution  $\mathbf{w} \sim \text{Dir}(\alpha)$ , which lands on  $\Delta^2$  by construction, so every draw is a valid weighting with no grid or renormalization step. The concentration  $\alpha$  sets the coverage pattern:  $\alpha = (1, 1, 1)$  samples the triangle uniformly (no prior preference); large  $\alpha$  concentrates draws near the equal-weight centroid (a local sensitivity check); small  $\alpha$  pushes them toward the corners (single-axis stress tests). The selection’s sensitivity to  $\mathbf{w}$  is then reported. This turns “how independent is RRM of accuracy?” from a fixed property into a tunable one: the weight slides the metric between an accuracy proxy (mass on  $v_0$ ) and pure reliability (mass off it), letting one choose the weighting that maximizes added information while still selecting good models.

**Roadmap.** Beyond RRM the suite includes weight-free global comparisons (rank-aggregation and Pareto-front selection over accuracy and Bayesian reliability; BEARDOWN App. B) and two candidate fourth axes  $v_3$ : a false-positive/false-negative–structure term, and, for the ATLAS reproduction specifically, an ABCD background-closure-quality term. All are exploratory and not claims of the main proposal. The configuration this procedure selects is the author-optimized variant (model 2) delivered for each reproduction in Section 3b.

## D. Model Roster — All Ten Builds

Values below are read from the configuration files in the experiments repository ([INFO\\_698\\_experiments](#)), which is the source of truth: `projects/<domain>/configs/`. Where a run overrode a config default, the run value is given in Section 7 and the difference is noted here. Ten models were built — three on the first pass in each domain, plus phonon mk4 after the variance lock; one per domain is preloaded in the observatory (§7.1).

### D.1 Genre — BEARDOWN family (GTZAN, $K = 10$ )

Shared across all three: stratified split, seed 42; 57 tabular features (`length` dropped, faithful to `cfg_14`); mel recipe `sr = 22050`,  $n_{\text{fft}} = 2048$ , hop 512,  $n_{\text{mels}} = 128$ , image 128 px; 3-fold CV; head remains deterministic, with LLLA applied post-hoc as the evaluation lens rather than as a swept axis.

|                                    | Identity                                                                                                                                                     | Architecture                                                                                                                                                                    | Training                                                                                                                                                                                                         |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>mk1</b><br><i>beardown</i>      | Faithful <code>cfg_14</code> reproduction, no sweep. Representation <code>fused</code> : 30 s tabular $\bowtie$ per-track mel image.                         | Spec CNN conv $[[32,3],[64,3],[128,3]]$ , embed 512, dropout 0.188; tab MLP $57 \rightarrow [192]$ , embed 192, dropout 0.191; gated fusion; head penultimate 512, dropout 0.2. | Adam, lr $7.37 \times 10^{-5}$ , batch 8, 75 epochs, weight decay 0. Early stop on val accuracy, patience 4, restore best; ReduceLROnPlateau 0.5, patience 2, min $10^{-6}$ . Target 0.766, accept $\geq 0.75$ . |
| <b>mk2</b><br><i>beardown_rrm</i>  | Broad random sweep over the same deterministic pipeline, selected by the RRM reliability metric of Appendix C rather than raw accuracy.                      | mk1 as base; the sweep overrides the fields named in its <code>sweep.space</code> . Deployed winner <code>cfg_17</code> : image branch 768, tabular 256, concatenative fusion.  | As mk1 except where swept. Selection driven by the posterior of the final layer — the $U$ term of RRM.                                                                                                           |
| <b>mk3</b><br><i>beardown_3sec</i> | The data-limited control. Identical sweep to mk2 in every axis except the data: 3 s segments, $\sim 10$ rows per track, representation <code>fused3</code> . | Base is the mk2 winner ( <code>cfg_17</code> ). Each 3 s segment joins many-to-one to its parent track’s single mel image.                                                      | Split on <i>track</i> ids so all segments of a recording land in one split — a leakage guard, not a convenience. Tests whether $\sim 10\times$ the rows revive the epistemic axis.                               |

**Table 21:** Genre models *mk1*–*mk3*.

## D.2 Phonon — E(3)-equivariant family (DFPT phonon DOS, 51 bins)

|                                 | Identity                                                                                                                                              | Architecture                                                                                                                                                                                                | Objective / training                                                                                                                                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>mk1</b><br><i>e3nn_repro</i> | Faithful reproduction of Chen <i>et al.</i> (2021). Serves as the frozen trunk for mk4.                                                               | in = 118, embedding 64, out 51; 2 layers, mul = 32, $\ell_{\max} = 1$ , max radius 5.0 Å, 10 radial basis functions. Data: 1,524 materials on the authors' own split, 1220/152/152 train/test/val.          | MSE. 64 epochs, batch 1, lr $5 \times 10^{-3}$ , weight decay 0.05, $\gamma = 0.96$ . Target $\bar{\omega}$ within 10 % = 0.70.                                                                                                                                         |
| <b>mk2</b>                      | mk1's architecture with an information-theory-aware objective: MSE is blind to the fact that the 51 bins lie on a metric frequency axis.              | Unchanged from mk1.                                                                                                                                                                                         | $L = (1 - \alpha)\text{MSE} + \alpha\text{DIST} + \beta \Delta H $ , DIST = 1-D Earth Mover (closed form via cumsum) or Jensen–Shannon; $\alpha = 0.4$ settled. Early stopping on val frac@10 %.                                                                        |
| <b>mk3</b>                      | Objective sweep: which information-theoretic or thermodynamic term best places phonon mass?                                                           | Unchanged from mk1.                                                                                                                                                                                         | Grid over {term $\times$ weight}. Blend terms: EMD, JS, KL forward, KL reverse, cross-entropy. Additive terms include vibrational entropy $S_{\text{vib}}(T)$ . Winner retrained full length.                                                                           |
| <b>mk4</b><br><i>FORGE head</i> | Built after the variance lock. Varies the <i>likelihood and the head</i> , not the objective, on a frozen mk1 trunk, so any difference is the head's. | Scalar-only readout ( $\ell = 0$ , preserving equivariance). Config rank 16 = 2 planted + 6 $C_v$ modes + 8 shape; $T$ reported at 30/77/150/300/600/1000 K, floor 28.8 K. Boards in §7 run rank 32 and 51. | Gaussian likelihood, $\sigma^2$ from a representation floor computed from labels only, so bias contamination is structurally impossible; 6-bin taper for the banded variant. $\tau = 10^k \lambda_{\max}$ , $k \in [-4, 0]$ , 41 points, optimum by MacKay fixed point. |

**Table 22:** Phonon models mk1–mk4. mk4 is the fourth build described in §7.1.

**mk4 head roster.** H3 planted-banded is the production pick; H2 planted-diagonal is the diagonal reference and H6 PCA-diagonal the control (after Ben Mahmoud); then H1 direct-51, H4 softmax-51 multinomial with  $n_{\text{eff}} = 118$ , H5 softmax-52 multinomial with an overflow bin, and H7 heteroscedastic with learned  $\sigma^2$  (a control expected to fail A-invariance — and it does, §7.2). Ranked on frac@10 %, PICP95, CRPS and A-invariance; EAR is an output, never the selector.

## D.3 ATLAS — LLP genealogy (Run-3 MSVtx, endcap and barrel)

All three rungs share one entry point so the genealogy differs *only* by objective: same data, same architecture, same schedule. Two networks, each hidden = (128, 128) with a Linear(128  $\rightarrow$  1) and sigmoid — which is precisely the head LLLA requires. Dropout 0.1;  $k = 5$  out-of-fold; 60 epochs; batch 256; Adam lr  $10^{-3}$ ; weight decay  $10^{-4}$ ; gradient clip 5.0; seed 7; quantile scaling fitted inside each fold; balanced class weights.

| Rung                             | Objective                                                                                            | What it isolates                                                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R1</b><br><i>vanilla</i>      | Binary cross-entropy alone.                                                                          | The reproduction baseline. NN1 OOF AUC 0.8785 against a $\sim 0.86$ reference.                                                            |
| <b>R2</b><br><i>disco</i>        | $BCE + \lambda \cdot dCor^2$ , $\lambda = 5.0$ , after Kasieczka <i>et al.</i> (2021).               | Whether enforcing global decorrelation of the two discriminants is sufficient for ABCD closure. It is not (§7.4).                         |
| <b>R3</b><br><i>disco_dncnqn</i> | R2 plus the DNCn_QN closure term, $\mu = 1.0$ by default, $\mu \in \{2, 5\}$ at the operating point. | Whether a term aimed at the sparse high–high corner does work the global penalty cannot. It does: threefold better closure at fixed dCor. |

**Table 23:** ATLAS rungs R1–R3. Default  $\lambda = 5.0$ ,  $\mu = 1.0$ ; the replicated factorial of §7.4 sweeps  $\lambda \in \{2, 3, 5\}$  and  $\mu \in \{0, 1, 2, 5\}$  over three seeds.

## E. The Accuracy-versus-Signal Frontier (phonon)

The frontier behind the operating point quoted in §7.2. It is reported here rather than in the body because the conclusion — that accuracy, exposable epistemic signal and calibration move together along  $\tau$ , so the operating point is a choice on a measured curve — is stated in §7.2 and argued in §8, and does not need the full sweep alongside it.

mk4 exists for one reason: to make phonon readable by the FORGE four metrics (FF1–FF4, §5b). That needs a declared likelihood and a linear map for the posterior to sit on, and mk1 has neither. The head supplies both — and scores below mk1.

Four pieces of vocabulary carry the rest of this section. The **head** is the small linear layer fitted on top of the frozen mk1 network; it is the only place the posterior lives. Its **declared likelihood** is the stated noise model — the assumption about how a prediction can be wrong — and without one there is no irreducible term to separate out. **Tau** ( $\tau$ ) is the prior scale, the single knob governing how far the head’s weights may move from the trained values: large  $\tau$  pins them, small  $\tau$  lets them spread. **Rank** and  $d$  are the size of the head — how many directions it keeps, and how many parameters the posterior therefore covers.

The trade is systematic rather than one bad configuration. Sweeping  $\tau = 10^k \lambda_{\max}$  across seven decades moves accuracy and exposable epistemic signal in opposite directions, and moves calibration with the signal rather than the accuracy: across the same sweep P1CP95 — how often the 95 % interval really does contain the truth, where 0.95 is ideal — improves from 0.874 to 0.889, and the PIT Kolmogorov–Smirnov statistic, the distance of the calibration curve from ideal, falls from 0.256 to 0.156. Both, and every other metric named in this section, are defined in the glossary (§13).

| $k$            | +2    | +1    | 0            | −1    | −2    | −3    | −4           | −5    |
|----------------|-------|-------|--------------|-------|-------|-------|--------------|-------|
| frac@10%       | 0.711 | 0.711 | <b>0.717</b> | 0.691 | 0.684 | 0.678 | <b>0.664</b> | 0.671 |
| reducibility % | 0.00  | 0.00  | 0.01         | 0.09  | 0.36  | 1.14  | 2.52         | 3.92  |

**Table 24:** The phonon frontier across seven decades of the prior scale  $\tau = 10^k \lambda_{\max}$ . Accuracy and exposable epistemic signal move in opposite directions, which is what makes the operating point a choice on a measured curve rather than a target hit or missed.

**How to read it.** Left to right is the prior loosening. On the left the head is pinned to the trained weights, so it predicts almost exactly as mk1 does — and reducibility reads 0.00 %, meaning it has nothing to say about what more data would fix. On the right the weights are free to spread: accuracy gives up about five materials in 152, and the reducible share climbs to 3.9 %. *frac@10%* is the published accuracy metric — the fraction of test materials whose mean phonon frequency lands within 10 % of the truth — and *reducibility* is the share of the remaining doubt that more data could remove. The two move against each other. That is the trade, and two operating points fall out of it.

- **REPORTED** —  $k = 0$ , rank 51,  $d = 2601$ : frac@10% **0.717**, above both mk1 (0.711) and the published 0.70; reducibility 0.01 %, EAR −38.24 dB. The posterior has collapsed to a point mass and the instrument reads nothing.

- **INSTRUMENT** —  $k^* = -4.97$  by MacKay evidence, rank 32,  $d = 1632$ : frac@10% **0.664**, reducibility 2.5–3.9%, PICIP95 0.886, PIT-KS 0.149 on the shipped H3 head. Chosen with no test data.

Precision matters here, and so does what the fractions are fractions *of*. Scoring runs on Chen *et al.*'s own held-out split — 152 of the 1,524 materials, taken from their published index file, which is what makes the 0.70 benchmark comparable at all. On those 152, 0.664 against 0.711 is 101 materials correct versus 108: a gap of seven, roughly  $1.2\sigma$ , statistically indistinguishable but not the same number. And 0.664 sits *below* the published 0.70. Why the readable head cannot hold 0.711 at a usable  $\tau$  is not established, and is carried into §8 as open work.

The dashboard defaults to  $k^*$  rather than to the more flattering  $k = 0$  because FF3 says so: prior-leak reads 0.951 at  $k = 0$  against  $\sim 0.000$  at  $k^*$ , so almost all of the apparent epistemic variance at the accurate end came from the prior rather than from the data.

## F. The Deployed Site — Structure and Request Path

The observatory is served from [forge-observatory.com](https://forge-observatory.com), the deploy repository. Its rule is that nothing is developed here: an app matures through its mk cycle in INFO\_698\_experiments, and only a locked version is promoted into apps/. The structure below is the consequence of that rule rather than a filing convention.

```
forge-observatory/
|-- CNAME                forge-observatory.com
|-- index.html           splash -> model deck
|-- experiments.html     the deck
|-- coming-soon.html     interstitial for unbuilt plates
|-- docker-compose.yml   services + the tunnel connector
|-- Dockerfile           one image, CPU torch wheel pinned
|-- .env.example         TUNNEL_TOKEN goes here, never committed
|
|-- apps/                PROMOTED apps only
|   |-- genre/           app/ configs/ data/ eda/ src/
|   |-- phonon/          app/ configs/ data/ eda/ src/
|   '-- atlas/           (reserved)
|
|-- _shared/             schema, profiler, splits, eda, usage
|-- assets/              images, brand
|-- models/              promoted bundles
|
|-- publish_mk3.ps1      engine -> build -> smoke -> tunnel -> push
|-- RUNBOOK.md           first bring-up, P1 tunnel / P2 apex
'-- DROP_NOTES_s3..s8.md the promotion log, one file per drop
```

**How a request reaches a model.** The apex [forge-observatory.com](https://forge-observatory.com) is served by GitHub Pages from the documentation repository, so the splash and the deck are static and cost nothing to keep up. Each experiment lives on its own subdomain — genre., phonon., atlas. — resolved by a Cloudflare tunnel to the home box, where the connector forwards to the Compose service on its internal port (genre and phonon are published on 127.0.0.1:5601 and 5602, deliberately clear of the crowded 5000 range). Nothing is exposed by opening a port on the router; the connector dials out. That is what makes the \$0 hosting figure of \$5a a property of the architecture rather than a temporary arrangement.

**Publishing** is one command. `publish_mk3.ps1` starts the engine, builds and raises the Compose stack, runs a container smoke test against `/api/forge/meta`, brings up the tunnel and pushes. The smoke test is there because of a real failure: the first public build reached a remote machine and then died on `No module named 'torch'` — the image had never contained it. A hollow image can no longer pass publish.

**The design tier** is separate again. [FORGE\\_site\\_dev](#) holds the splash, deck, per-model and dashboard pages as `index/dash1/dash2/fidget/for`.html with every value stubbed, plus `boot_site_mk1.ps1`, which serves the folder it sits in on the first free port in 8000–8020. Interface work therefore never waits on a model, and a layout change cannot break a reproduction.

## 13. Glossary

Prepared by Nathan Herling

Terms are defined here on the understanding that a reader may meet them out of order. Where a term carries a measured value in this project, the value and the section that argues it are given with the definition.

| Term                                                   | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MCP</b><br><i>Model Context Protocol</i>            | An open standard for advertising software functions to a large-language-model client, so that the client can discover them and call them as tools rather than a human driving a user interface. FORGE’s S1 stretch (Table 5) would have published the compute functions this way — a manifest over the mathematics the dashboard already runs, adding no new science, which is why the proposal names it first in the descope order (§3b). Not attempted; see §5a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>LLLA</b><br><i>Last-Layer Laplace Approximation</i> | A Gaussian posterior fitted to the final linear layer of a trained network, with everything upstream frozen at its trained values. Closed form, so it recomputes in milliseconds, and sufficient to remove the asymptotic overconfidence of a ReLU network (Kristiadi <i>et al.</i> , 2020). It sees only the last layer — 1,632 of 2.4 M parameters on phonon — which is why model bias never enters the aleatoric term.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>HMC</b><br><i>Hamiltonian Monte Carlo</i>           | A gradient-informed sampler that draws from the true posterior rather than approximating it (Neal, 2011), usually in the self-tuning NUTS form (Hoffman and Gelman, 2014). In FORGE it is the referee for the fast path: comparing the two is what establishes whether the approximation can be trusted. On a linear-Gaussian head the posterior is exactly Gaussian, so the sampler agrees by construction and verifies the pipeline rather than the split.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>DNCn_QN</b>                                         | The closure term introduced in this project (§6.4). It penalizes the ABCD prediction error at the cut in units of that error’s own statistical noise, with a hinge so the penalty is silent below one standard deviation. Aimed at the sparse high–high corner where ABCD closure actually lives, which is the work a global decorrelation penalty cannot do.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>RRM</b><br><i>Robust Ranking Metric</i>             | A cohort-relative model selector (Appendix C): each configuration is scored by a penalty vector of inaccuracy, fold instability and posterior spread, and ranked by $1 - \ \mathbf{v}\ $ . Higher is better. It is undefined for a single model, which is why it requires a sweep. Used for selection only, never for reporting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>MacKay evidence fixed point</b>                     | The rule that sets the prior scale $\tau$ without using held-out labels. The <i>evidence</i> is the marginal likelihood $p(\mathcal{D} \mid \tau)$ , obtained by integrating the weights out of the posterior rather than fixing them at one value; maximizing it gives the fixed-point iteration $\tau \leftarrow \gamma(\tau) / \ \mathbf{w}\ ^2$ , where $\gamma(\tau) = \sum_i \lambda_i / (\lambda_i + \tau)$ is the effective number of parameters the data actually constrains — directions with $\lambda_i \gg \tau$ count fully, directions with $\lambda_i \ll \tau$ barely at all. Iterating to convergence returns $\tau^*$ . Because only training data enters, the operating point is chosen without touching the test set; on phonon the resulting $k^* = -4.97$ lands on the accuracy plateau (Appendix E), so the value found blind agrees with the value a labelled search would have found. After MacKay (1992). |

## Equations

Three numbered equations carry the project’s base measure. Their provenance differs and is stated with each.

**Equation (1) — the variance decomposition.**  $\text{Var}(y_k \mid x) = \mathbb{E}_\theta[\text{Var}(y_k \mid x, \theta)] + \text{Var}_\theta(\mathbb{E}[y_k \mid x, \theta])$ . The total variance of the prediction at resolving-axis index  $k$ , split into an *aleatoric* term  $A_k$  — the average spread the model still expects once the weights are fixed, which more data cannot remove — and an *epistemic* term  $E_k$ , the spread of the model’s own mean prediction as the weights range over the posterior, which more data can. *Provenance*: not original. The identity is the law of total variance, classical; conditioning it on  $\theta$  to separate the two kinds of uncertainty is standard in Bayesian deep learning and is operationalized for deep networks by Kendall and Gal (2017). What is original here is adopting it as FORGE’s base measure in place of entropy and mutual information, and reporting it per  $k$  rather than as a scalar (§6.1).

**Equation (2) — EAR, the epistemic-to-aleatoric ratio.**  $\text{EAR}_k = 10 \log_{10}(E_k/A_k)$ , in decibels. Positive values mean the

reducible part dominates — more or better data should help; negative values mean the irreducible part dominates and the model is at its limit. Aggregated as a ratio of sums, never a mean of ratios, which diverges as  $A_k \rightarrow 0$ . *Provenance*: an adaptation, and claimed as original in that form. The  $10\log_{10}$  decibel convention for a ratio of powers is standard signal-processing practice and is borrowed; its use as an uncertainty readout is introduced here. The adaptation is defensible rather than decorative because under Equation (1) the argument  $E_k/A_k$  is a ratio of variances, which is literally a signal-to-noise ratio — so the decibel is derived, not analogy. No prior use of the epistemic-to-aleatoric ratio expressed in decibels was found.

**Equation (3) — reducibility.**  $\text{reducibility}_k = E_k/(E_k + A_k)$ , the share of the budget at  $k$  that is reducible, on  $[0, 1]$ . *Provenance*: not original and claims nothing — an ordinary proportion-of-variance share. It is reported alongside EAR because it is bounded and reads directly as a percentage, where EAR is unbounded and reads better on a dial.

## Domain terms

| Term                                     | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High-energy physics</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>LLP</b><br><i>long-lived particle</i> | A hypothesized particle that travels a measurable distance before decaying, so its decay products originate away from the collision point. The search looks for that displacement rather than for the particle itself.                                                                                                                                                                                                                                                                       |
| <b>MSVtx</b>                             | Muon-spectrometer vertex: a decay reconstructed in ATLAS's outermost detector layer. The Run-3 MSVtx analysis (ANA-EXOT-2025-24) is the search this reproduction targets.                                                                                                                                                                                                                                                                                                                    |
| <b>endcap, barrel</b>                    | The two geometric regions of the detector — barrel wrapping the beamline, endcaps capping it. They are analysed separately because acceptance and background differ. Endcap carries 24,850 background events here against the barrel's 2,280, which is why only endcap results are claimed (§7.4).                                                                                                                                                                                           |
| <b>signal region, control region</b>     | The cut-defined box where signal is expected, and the three boxes used to predict its background under ABCD.                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>BCE</b>                               | Binary cross-entropy, the classification loss the DisCo and closure penalties are added to.                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Materials science</b>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>DOS</b><br><i>density of states</i>   | How many vibrational modes a crystal has at each frequency — the 51-bin spectrum the phonon model predicts. Its shape determines thermal properties.                                                                                                                                                                                                                                                                                                                                         |
| <b>DFPT</b>                              | Density-functional perturbation theory, the first-principles calculation that produced the ground-truth spectra (Petretto <i>et al.</i> , 2018).                                                                                                                                                                                                                                                                                                                                             |
| <b>E(3)-equivariant</b>                  | A network whose output transforms correctly when the input crystal is rotated, translated or reflected, rather than having to learn those symmetries from data. It is also why the network has no linear last layer (§5a).                                                                                                                                                                                                                                                                   |
| <b>zero-point energy, heat capacity</b>  | Two physical quantities computable directly from the DOS. FORGE <i>plants</i> them in the readout basis so each carries its own posterior rather than being derived afterwards.                                                                                                                                                                                                                                                                                                              |
| <b>Audio</b>                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>GTZAN</b>                             | The standard ten-genre music benchmark, 100 thirty-second clips per genre. Its label noise is real and sits outside the aleatoric term, so the genre floor reads low.                                                                                                                                                                                                                                                                                                                        |
| <b>mel spectrogram</b>                   | Audio rendered as an image — time against frequency on a perceptual scale — which is what the CNN branch reads.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Data science</b>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Bayesian vetting</b>                  | Judging a trained model by its posterior rather than by a single score: reading the variance budget of Equation (1) along the resolving axis, checking the fast split against the sampler (FF2), and refusing a head that fails the aleatoric referee even when its coverage is nominal — the H7 case of §7.2. Accuracy and calibration are inputs to the verdict, not the verdict.                                                                                                          |
| <b>MaxEnt background</b>                 | Maximum-entropy background estimation: per-feature one-dimensional maximum-entropy marginals fitted to the total background, whose product is the factorized joint the ABCD method already assumes, with copula rungs adding cross-moments between the NN1 and NN2 feature blocks. Samples drawn from that joint and pushed through the frozen discriminants reconstruct the signal-region count independently of the ABCD arithmetic. Companion work to §6.4; not exercised in this report. |
| <b>ABCD validation</b>                   | The gate a decorrelated pair of discriminants must clear before its background estimate is trusted: distance correlation below threshold, small median non-closure $ nc _{\text{med}}$ , and a stated fraction of evaluation cells within $2\sigma$ of their prediction — the three columns of Table 19. The gates are not monotone together, which is why §7.4 reports all of them rather than one.                                                                                         |



## Instrument terms

| Term                                            | Meaning                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>trunk</b>                                    | The trained network below the readout head. Frozen throughout: nothing FORGE does changes it, which is what makes the instrument a measurement of the model rather than a modification of it.                                                                                                                                                                                               |
| <b>bundle</b>                                   | The export format the observatory reads — last-layer weights, pre-projected curvature, the cached feature matrix $\varphi$ , and a scalar record. One bundle per model; eight on the board.                                                                                                                                                                                                 |
| <b>the contract</b>                             | The tagged scalar record inside every bundle, schema <code>forge-metrics-1.0</code> : 46 fields, each marked universal, family-specific or conditional. A null conditional field is information — “FF2 not applicable, linear-Gaussian head” is a reading, not a gap — and the tagging is what lets the dashboard refuse an invalid cross-family comparison instead of silently making one. |
| <b>MAP</b>                                      | Maximum a posteriori — the single best weight setting, the point estimate a posterior is fitted around.                                                                                                                                                                                                                                                                                     |
| <b>conjugate</b>                                | A prior and likelihood whose combination has a closed-form posterior. A linear-Gaussian head is conjugate, which is exactly why HMC cannot disagree with LLLA there and FF2 reads nothing on phonon.                                                                                                                                                                                        |
| <b>representation floor</b>                     | A noise level computed from the labels alone, never from the model's residuals. Because it never sees the model, bias cannot leak into the aleatoric term — which is what H5 has and H7 does not.                                                                                                                                                                                           |
| <b>softmax, sigmoid, multinomial, Bernoulli</b> | The likelihood families in play: softmax/multinomial for the ten genre classes and the 51 DOS bins, sigmoid/Bernoulli for the binary ATLAS discriminants. Neither is conjugate, which is why FF2 would inform on those two and not on phonon.                                                                                                                                               |
| <b>seed-median</b>                              | A cell reported as the median over repeated training runs differing only in random seed, rather than as one run. The ATLAS factorial uses three seeds per cell.                                                                                                                                                                                                                             |

## Metrics and tests

| Name                           | What it measures                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>frac@10%</b>                | The published phonon accuracy metric: the fraction of test materials whose predicted mean phonon frequency $\bar{\omega}$ lands within 10 % of the true value. Chen <i>et al.</i> report 0.70 on their own held-out split of 152 materials; the same split and metric are used here.                                                                                                                                                       |
| <b>PICP95</b>                  | Prediction-interval coverage probability at the 95 % level — how often the model's 95 % credible interval actually contains the truth. 0.95 is ideal; below it the intervals are too narrow (over-confident), above it too wide.                                                                                                                                                                                                           |
| <b>PIT, PIT-KS</b>             | The probability integral transform maps each observation through its own predictive CDF; if the predictive distribution is right, the results are uniform on $[0, 1]$ . PIT-KS is the Kolmogorov–Smirnov distance of that sample from uniform — lower is better. Unlike PICP95 it uses the whole distribution, not one interval, so it catches the shape of a miscalibration and not just its width.                                       |
| <b>A-invariance</b>            | The referee on the aleatoric term. Aleatoric variance is <i>defined</i> as the part that does not shrink with more data, so $A$ must not move with $N$ ; the test reports the drift as a percentage. A noise level read from the labels alone passes trivially, having never seen the model. A $\sigma^2$ fitted to the model's own residuals absorbs model bias into $A$ and must fail — that contrast is what the test exists to expose. |
| <b>CRPS</b>                    | Continuous ranked probability score: a proper scoring rule comparing the whole predictive distribution against the observed value, rewarding sharpness only when it is justified by accuracy. Used alongside frac@10 % and PICP95 to rank heads.                                                                                                                                                                                           |
| <b>dCor</b>                    | Distance correlation between the two ATLAS discriminants. Unlike Pearson correlation it detects non-linear dependence, and it is zero only under genuine independence — which is the condition the ABCD background estimate requires. A <i>global</i> statistic, dominated by the bulk of the sample.                                                                                                                                      |
| <b>ABCD, <math> nc </math></b> | ABCD estimates background in a signal region from three control regions, and is valid only if the two discriminants factorize. Non-closure $ nc $ is the relative error of that prediction at a given cut; $ nc _{\text{med}}$ is its median across the evaluation grid. Unlike dCor it depends on the sparse high–high corner, which is why the two can move independently (§7.4).                                                        |
| <b>OOF AUC</b>                 | Area under the ROC curve computed out-of-fold: every prediction is made by a model that did not see that row in training, so the figure is not inflated by the fit. The ATLAS reproduction reports NN1 and NN2 this way.                                                                                                                                                                                                                   |

## Symbols

| Symbol                      | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $k$                         | Index along the <i>resolving axis</i> — the subdivisions the model's own output already has, in place before FORGE reads anything: class for genre, energy bin $\omega$ for phonon, signal region for ATLAS (Table 15). Readings are reported per $k$ and only aggregated for a headline (§6.3).                                                                                                                                                                                                                    |
| $\phi$                      | The feature vector entering the readout head; $\phi_k$ is its $k$ -th component. For phonon this is the trained network's own 51-bin prediction (§5a).                                                                                                                                                                                                                                                                                                                                                              |
| $\Sigma$                    | Posterior covariance over the last-layer weights, $\Sigma = (H_{\text{GGN}} + \tau I)^{-1}$ . $\Sigma_{(k,k)}$ is its $k$ -th diagonal block.                                                                                                                                                                                                                                                                                                                                                                       |
| $H_{\text{GGN}}$            | Gauss–Newton curvature of the loss at the fitted weights — the second-derivative information the Laplace approximation is built from.                                                                                                                                                                                                                                                                                                                                                                               |
| $\tau$                      | Prior scale, the single regularization knob. Set by the MacKay evidence fixed point rather than tuned; quoted as $\tau = 10^k \lambda_{\text{max}}$ , so $k$ in that context is a decade exponent, not the axis index above.                                                                                                                                                                                                                                                                                        |
| $k$ (as an exponent), $k^*$ | $\tau$ is quoted on a log scale relative to the curvature it regularizes: $\tau = 10^k \lambda_{\text{max}}$ , with $\lambda_{\text{max}}$ the largest eigenvalue of the Gauss–Newton matrix, so $k$ is a decade exponent. $k^*$ is its value at the evidence optimum — on phonon $k^* = -4.97$ , roughly five decades below $\lambda_{\text{max}}$ . Distinct from the resolving-axis index $k$ above; context separates them, since one is an exponent on $\tau$ and the other a subscript on $y$ , $A$ and $E$ . |
| $\lambda_i, v_i$            | The $i$ -th eigenvalue and eigenvector of $H_{\text{GGN}}$ . Directions with $\lambda_i \gg \tau$ are pinned by data; those with $\lambda_i \ll \tau$ are held by the prior.                                                                                                                                                                                                                                                                                                                                        |
| $A_k$                       | Aleatoric variance at $k$ — the part that is irreducible, and would not shrink with more data.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $E_k$                       | Epistemic variance at $k$ — the reducible part, attributable to not having pinned the weights down. $E = \sum_k E_k$ .                                                                                                                                                                                                                                                                                                                                                                                              |
| $\text{EAR}_k$              | Epistemic-to-aleatoric ratio, $10 \log_{10}(E_k/A_k)$ , in decibels. A ratio of variances is literally a signal-to-noise ratio, so the decibel is derived rather than borrowed.                                                                                                                                                                                                                                                                                                                                     |
| $c_i$                       | The share of $E$ carried by eigendirection $i$ : $c_i = (\phi \cdot v_i)^2 / (\lambda_i + \tau)$ . Summed over the directions below $\tau$ it gives the prior-leak share.                                                                                                                                                                                                                                                                                                                                           |
| $\epsilon_k$                | Signed relative error of the fast split against the sampler, $(E_k^{\text{LLLA}} - E_k^{\text{HMC}})/E_k^{\text{HMC}}$ . The sign is kept: negative is the over-confident direction.                                                                                                                                                                                                                                                                                                                                |
| $a_k$                       | The share of $E$ allocated to feature $k$ , $a_k = \phi_k(\Sigma\phi)_k$ . Exact by Euler's theorem, so $\sum_k a_k = E$ with no residual. Plotted signed — anti-correlated features contribute negatively.                                                                                                                                                                                                                                                                                                         |
| PR                          | Participation ratio — a measure of how many features share the epistemic budget. Near 1 when the allocation $a_k$ is spread evenly across all of them, near $1/d$ when one feature carries it. Reported as concentration, $100(1 - \text{PR})$ , so a high number means the doubt is localized.                                                                                                                                                                                                                     |
| $\gamma(\tau)$              | Effective number of parameters the data constrains, $\sum_i \lambda_i / (\lambda_i + \tau)$ . The quantity the evidence fixed point balances against $\ w\ ^2$ .                                                                                                                                                                                                                                                                                                                                                    |

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# NTH\_INFO698\_final\_report

Final Audit Report

2026-08-10

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